

ORIGINAL ARTICLE

Estimation of dual permeability model parameters in agricultural fields using a genetic algorithm-based inverse modeling framework

Nirali Vashishth  | Anoop Pandey  | Richa Ojha 

Indian Institute of Technology Kanpur,
Kanpur, India

Correspondence

Nirali Vashishth, Indian Institute of
Technology Kanpur, Kanpur, India.
Email: nirali@iitk.ac.in

Assigned to Associate Editor Haiyun Shi.

Abstract

Macropores significantly influence water movement in agricultural fields. The dual permeability model (DPM) is generally used to simulate flow in macroporous soils. However, the parameters of DPM are poorly defined and difficult to measure, impeding its applicability at various spatiotemporal scales. This study proposes a genetic algorithm (GA)-based optimization framework to estimate DPM parameters from soil moisture observations, providing a cost-effective, non-intrusive field-scale alternative to tension-disc infiltrometer-based local-scale parameterization techniques. The robustness of GA is tested under varying atmospheric boundary conditions using surface and/or rootzone soil moisture observations. Additionally, the impact of temporal scales and noise in soil moisture data on parameter estimation is analyzed. Parameters are optimized for three distinct cases: (a) matrix parameters, (b) macropore parameters, and (c) combined matrix and macropore parameters. Parameters θ_{sm} and n_m are easily identified with relative error <3%, and K_{sm} and K_{sf} are estimated with relative error up to 17% and 12.5%, respectively. The findings of the optimization algorithm are validated with field measurements for a site at IIT Kanpur, showing similar trends as numerical experiments. Daily root zone soil moisture leads to reliable estimates of DPM parameters. The estimates of macropore parameters are sensitive to macropore fraction (ω_f). The implications of using DPM on soil moisture dynamics and crop water requirements are discussed. The results demonstrate the efficacy of the proposed algorithm in estimating DPM parameters.

Plain Language Summary

Water movement in soil is influenced by both small pores and larger channels called macropores, which traditional models often fail to fully capture. The dual

Abbreviations: DPM, dual permeability model; ET, evapotranspiration; GA, genetic algorithm; MAPE, mean absolute percentage error; NSE, Nash–Sutcliffe efficiency; RMSE, root mean square error; RZSM, root zone soil moisture; SHPs, soil hydraulic parameters; SPM, single porosity model; SSM, surface soil moisture; SWRC, soil water retention curve; TDI, tension disc infiltrometer; UI, uniqueness index.

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2025 The Author(s). *Vadose Zone Journal* published by Wiley Periodicals LLC on behalf of Soil Science Society of America.

permeability model (DPM) can be used to determine water movement in soils with large pores, but it is difficult to estimate its parameters. This study presents a new approach using genetic algorithms and soil moisture data to estimate DPM parameters. Daily root-zone soil moisture is most effective, especially under variable weather conditions. The study also compared DPM with simpler models and found that DPM more accurately simulated soil moisture, particularly at deeper depths, and better predicted irrigation needs. Results show that macropores influence both infiltration and water redistribution in the root zone. The proposed method has the potential to improve irrigation planning and better support larger-scale hydrological and land-surface models.

1 | INTRODUCTION

Modeling water movement in soil is crucial for understanding hydrological processes, managing agricultural systems, and optimizing irrigation practices. The simplest representation of the soil pore system is a single-domain model with homogeneous soil, where flow is governed by the Richards equation. However, this assumption may not hold true in agricultural fields, which are influenced by various natural and human-induced factors leading to soil heterogeneity and presence of fractures and macropores (Abu-hashim et al., 2023; Kodešová et al., 2012; Liu et al., 2016; Luo et al., 2008; Talukder et al., 2023).

Macropores enable preferential flow, rapid water movement through larger voids bypassing the soil matrix, making single-domain models inadequate (Beven & Germann, 2013; Nimmo, 2021). Field observations confirm the substantial role of macropores in contributing to preferential flow in agricultural land. For example, Cameira et al. (2003) observed that agricultural fields with minimum tillage can allow over 85% of total water flow to occur through macropores; Garg et al. (2009) observed 52% of the total flux in paddy field to be passing through meso- and macropores, while Cullum (2009) observed that macropores allowed as high as 55% bromide transport under no-tillage condition for a 5-h storm event. In addition to this, preferential flow pathways influence nutrient cycles, subsurface drainage, and soil structural properties (Franklin et al., 2021; Fuhrmann et al., 2019; Plier et al., 2020), underscoring their importance in crop water modeling.

In the last few decades, modeling studies have incorporated macropores in soils by dividing the soil media into two domains (matrix and macropore), generally known as dual-domain models (Chen et al., 1993; Choi et al., 1997; Djabbelkhir et al., 2017; Germann, 1990; Jarvis et al., 1991; Larsbo et al., 2005; Nimmo, 2010). Among the existing models, dual porosity model (van Genuchten & Wierenga, 1976) and dual permeability model (DPM) (Gerke & van Genuchten, 1993a) are frequently employed to simulate water movement

in macroporous soils. While these models offer a more realistic representation of flow in macroporous soils, the estimation of soil hydraulic parameters (SHPs) in them remains a primary concern. A full-fledged DPM requires 17 parameters to describe flow assuming advective transport in matrix domain (six parameters each to characterize the soil matrix (θ_{rm} , θ_{sm} , α_m , n_m , l_m , and K_{sm}) and macropores (θ_{rf} , θ_{sf} , α_f , n_f , l_f , and K_{sf}) and five parameters to define interface mass transfer (ω_f , K_{as} , a , β , and γ_w). The number of parameters increases as we consider source or sink terms such as root water uptake in the soil media (Pandey & Ojha, 2024).

The matrix parameters can be measured using standard laboratory or in situ experiments. However, macropore and interface mass transfer parameters are not directly measurable and must be estimated indirectly, often through inverse modeling (Arora et al., 2011; Das & Mohanty, 2006; Kodešová et al., 2010; Köhne et al., 2002, 2006; Peters & Durner, 2008; Ramos et al., 2006; Soracco et al., 2011; Valdes-Abellan et al., 2018). Inverse modeling studies at laboratory scale often link infiltration and drainage data to DPM parameters, thereby inversely estimating macropore and mass transfer parameters (Arora et al., 2011; Köhne & Mohanty, 2005; Köhne et al., 2006). Field studies have combined tension disc infiltrometers (TDI) (Faúndez Urbina et al., 2021), double ring infiltrometers or Guelph permeameters, dye tracers (Kodešová et al., 2010; Soracco et al., 2011), and tracer transport data (Köhne et al., 2006, 2009; Lennartz et al., 1999) to estimate macropore properties.

These studies have highlighted the importance of choice of sampling algorithm and the significance of matrix–macropore interface in flow experiments. However, there are some major challenges observed in these studies: (i) field measurements like pressure head, drainage fluxes, transport data, and the use of TDI for estimating DPM parameters are tedious, expensive, and time-consuming (Wang et al., 1998) and rarely available at large spatial scales; (ii) parameter estimates from small-scale experiments may not translate directly to heterogeneous field conditions; and (iii) the widely used

Levenberg–Marquardt optimization algorithm for inverse parameter estimation often converge to local minima in complex multimodal parameter spaces, leading to suboptimal solutions (Nocedal & Wright, 2006; Salonga et al., 2019). These limitations elucidate why DPM remains underutilized despite its importance in agricultural applications, forming the motivation for this work.

As measurement techniques have advanced, in situ soil moisture measurements and remote-sensing-based soil moisture observations are readily available with higher accuracy at various spatial and temporal scales (Dorigo et al., 2011, 2017; Ochsner et al., 2013; Walker et al., 2004). Soil moisture observations have now become a key source to infer the conditions conducive for preferential flow (Kang et al., 2023; Kulasekera et al., 2011; Nimmo et al., 2025; Oberdörster et al., 2010). While Ines and Mohanty (2009) and Charoenhirunyingyos et al. (2011) have used either near-surface remotely sensed or in situ soil moisture data for estimating effective SHPs of single porosity model, its potential for estimating macropore parameters has not been explored. Furthermore, genetic algorithms (GAs) offer an alternative to gradient-based methods in these inverse problems (Ines & Droogers, 2002; Ines & Mohanty, 2008; Sonkar et al., 2018, 2019), as they efficiently explore a large solution space and are less likely to get stuck in local minima, albeit at higher computational cost (Gad, 2023; Mitchell, 1998).

Previous studies have suggested that macropores can contribute to water movement not only during infiltration events but also during redistribution phase in root zone (Beven & Germann, 2013; Jarvis, 2020). Based on this, we hypothesize that surface and/or root zone soil moisture (SSM and RZSM) at different temporal scales retains signatures of macropore flow. These signatures can be used to estimate macropore parameters with inverse modeling. Till now, soil moisture measurements at different temporal resolutions have not been used to estimate macropore flow model parameters numerically or experimentally.

The specific objectives of this study are to (a) develop a GA-based optimization approach to estimate parameters of DPM using soil moisture observations as a search criterion, (b) analyze the robustness of GA in parameter estimation under varying atmospheric boundary conditions at different temporal scale (daily and weekly) and in presence of noise in soil moisture observations, (c) understand how parameter interdependence affects the GA's performance, (d) validate the proposed framework using field observations, and (e) analyze the implications of using DPM over a single porosity model in simulating water movement and estimating crop water requirement.

The remainder of this paper is organized as follows. The methodology for the study is described in Section 2. Section 3 presents the field experiments. The results are given in Section 4. The implications of using DPM over single-porosity

Core Ideas

- A search algorithm is proposed to inversely estimate the parameters of dual permeability model (DPM) using soil moisture.
- The DPM parameters are estimated for various numerical and experimental conditions.
- Reliable estimate of macropore fraction increased the estimation accuracy of other macropore parameters.

model (SPM) are discussed in Section 5. The conclusions of the study are presented at the end.

2 | MATERIALS AND METHODS

To achieve the objectives of this study, numerical experiments were conducted using a hypothetical parameter set derived from Šimůnek et al. (2003). In these experiments the soil domain was subjected to varied atmospheric boundary conditions. The GA used for inverse optimization was tuned, and its robustness was analyzed for different parameter bounds and combinations. Field experiments were then carried out to validate the proposed framework under real-world conditions. The validation involved comparing the model's output with observed data to assess its applicability and accuracy.

The following sub-sections describe the DPM, GA-based optimization framework, and numerical experiments.

2.1 | Dual permeability model

Conceptually, the DPM assumes fast water flow in one region (macropore domain) when close to full saturation and slow in the other region (matrix domain). The flow in these two domains is defined by the Richards equation (Equations 1 and 2, respectively), but with a different set of hydraulic parameters (Gerke & van Genuchten, 1993a; Šimůnek et al., 2003). In this study, Hydrus-1D (Šimůnek et al., 2013) was used for the simulations of DPM.

$$\frac{\partial \theta_f}{\partial t} = \frac{\partial}{\partial z} \left(K_f \frac{\partial h_f}{\partial z} + K_f \right) - S_f - \frac{\Gamma_w}{\omega_f} \quad (1)$$

$$\frac{\partial \theta_m}{\partial t} = \frac{\partial}{\partial z} \left(K_m \frac{\partial h_m}{\partial z} + K_m \right) - S_m + \frac{\Gamma_w}{1 - \omega_f} \quad (2)$$

Where, subscripts f and m denote macropore and matrix domain, respectively. The vertical coordinate, z , is positive upward [L], t is time [T], h is the pressure head [L], θ is the

water content [L^3L^{-3}], K is the unsaturated hydraulic conductivity [LT^{-1}], Γ_w is the rate of water exchange between the two domains [T^{-1}] and ω_f is the volume fraction of macropore (V_f) relative to total soil volume (V), as given in Equation (3):

$$\omega_f = \frac{V_f}{V} \quad (3)$$

For making the model less extensive in terms of number of parameters, a simple first order mass transfer between the soil matrix and macropores is assumed, which is proportional to difference in pressure heads between two pore regions (Arora et al., 2011; Gerke & van Genuchten, 1993b) and is given as follows:

$$\Gamma_w = \alpha (h_f - h_m) \quad (4)$$

where α is a first order mass transfer coefficient, which can be defined for well-defined geometries as follows:

$$\alpha = \frac{\beta}{a^2} K_a \gamma_w \quad (5)$$

where β is shape factor that depends on the geometry, a is characteristic length of aggregate, γ_w is scaling factor, and K_a is the effective hydraulic conductivity of matrix–macropore interface. The effective hydraulic conductivity of the matrix–macropore interface is described using a simple arithmetic average involving both h_m and h_f given by Gerke and van Genuchten (1996) Equation (6)

$$K_a(h) = 0.5 (K_a(h_f) + K_a(h_m)) \quad (6)$$

2.1.1 | Constitutive relationships

Explicit relationships are used between the dependent variable (h) and the nonlinear terms (θ and K) (Mualem, 1976; van Genuchten, 1980) for both domains.

$\theta - h$ correlation:

$$\theta(h) = \omega_f \theta_f(h) + (1 - \omega_f) \theta_m(h) \quad (7)$$

$$S_{e,i} = \begin{cases} [1 + (\alpha_i h)^{n_i}]^{-m_i} & \text{for } h \leq 0 \\ 1 & \text{for } h > 0 \end{cases} \quad (8)$$

$$\theta_i(h) = \theta_{r,i} + (\theta_{s,i} - \theta_{r,i}) S_e \quad (9)$$

where $i = 1$ indicates matrix and 2 denotes macropore domain; $m_i = 1 - 1/n_i$, and S_e is the effective saturation; and α_i , n_i ,

$\theta_{r,i}$, and $\theta_{s,i}$ are soil water retention curve parameters for i th domain.

$K - \theta$ correlation:

$$K(h) = \omega_f K_f(h) + (1 - \omega_f) K_m(h) \quad (10)$$

$$K_i(h) = \begin{cases} K_{s,i} S_{e,i}^{l_i} \left[1 - (1 - S_{e,i}^{1/m_i})^{m_i} \right]^2 & \text{for } h \leq 0 \\ K_{s,i} & \text{for } h > 0 \end{cases} \quad (11)$$

where $K_i(h)$ is the hydraulic conductivity of i th domain, $K_{s,i}$ is the saturated hydraulic conductivity for i th domain, and l_i is the pore connectivity parameter.

2.1.2 | Root water uptake model

The sink term in the Richards equation ($S(h)$ in Equation [1] and [2]) is defined here as volume of water removed from a unit volume of soil per unit time due to plant root water uptake. Root water uptake is simulated using Ojha and Rai (1996) (O-R model). The variability in root density and root distribution with depth within the root zone indicates that a non-linear model would be more appropriate for simulating water extraction by roots. The nonlinear model developed by Ojha and Rai (1996) considers maximum water extraction at the root's top and minimum at the bottom, reflecting the nonlinear variation of root density with soil depth. Thus, this study incorporates O-R model for root water uptake. The potential rate of water extraction by plants from the soil can be expressed as follows:

$$S_{\max} = b_x T_p \quad (12)$$

where T_p is potential rate of transpiration; b_x is root distribution function, characterized as follows:

$$b_x = \frac{(\beta + 1)}{z_{r\max}} \left(1 - \frac{z}{z_{r\max}} \right)^\beta \quad (13)$$

where z is the depth below soil surface, and $z_{r\max}$ is the maximum depth of root zone, β is the nonlinear root water uptake parameter, which has been empirically estimated using following expression (Kumar et al., 2015; Shankar et al., 2012):

$$\beta_{\text{optimal}} = 5.1128 T_s^2 - 6.117 T_s + 3.1545 \quad (14)$$

where T_s refers to non-dimensional plant physiological parameter, specific transpiration, which is defined as follows:

$$T_s = \frac{T_{j\max}}{z_{r\max}} \times t_{\text{peak}} \quad (15)$$

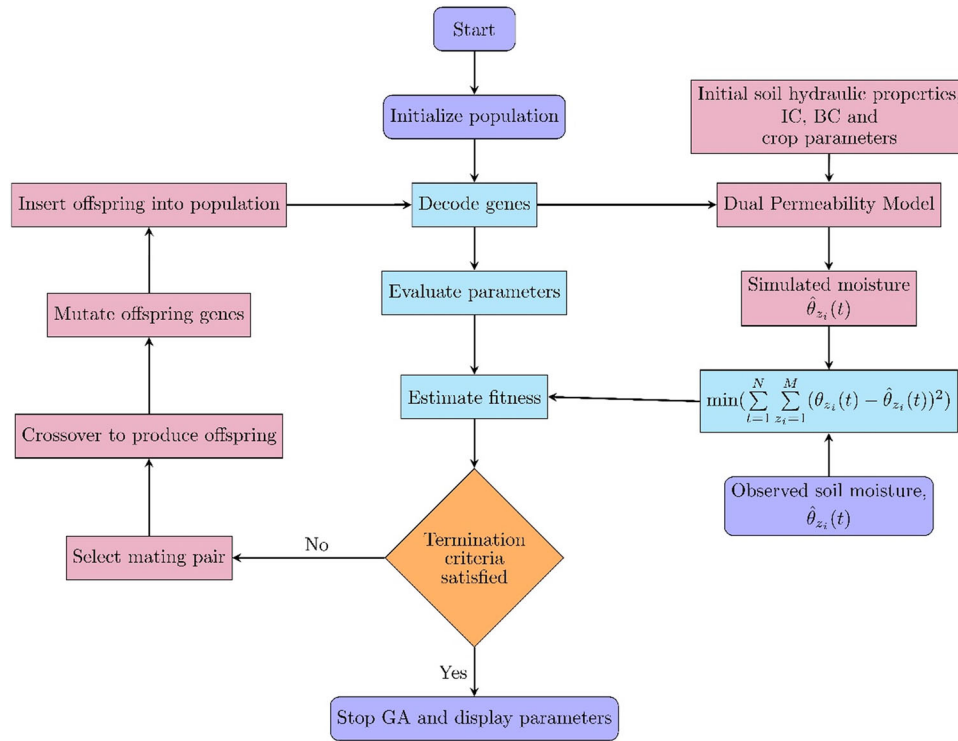


FIGURE 1 Flowchart showing the inverse modeling framework using genetic algorithm.

where T_{jmax} is the maximum daily transpiration, z_{rmax} is the maximum root depth and t_{peak} is the time to attain maximum transpiration.

2.2 | GA-based optimization framework

The inverse modeling framework used in this study is shown in Figure 1. The present study employed open-source python library, PyGAD for optimization. GA mainly involves four processes: (i) fitness criteria, (ii) selection, (iii) crossover, and (iv) mutation. An initial population is generated within the given range of parameter values and fitness of initial population is evaluated using fitness function. The initial population undergoes selection, cross-over and mutation and thus gets evolved. Evolution starts by selecting the fittest parents and further employing mutation and crossover to improve the parents' capabilities. By mating the best parents and producing more offsprings, a better solution is likely to be found after each generation. This process is repeated for a limited number of generations. Detailed description of GA can be found in Goldberg (1991).

The fitness function ($F(p)$) for inverse modeling was defined as follows:

$$F(p) = \frac{1}{f_k(p) + 10^{-5}} \quad (16)$$

where $f_k(p)$ is the objective function, which minimizes sum of square of the difference of simulated and observed soil moisture at different depths in the root zone (i.e., 10, 25, 50, and 80 cm in this study).

$$f_k(p) = \min \left(\sum_{t=1}^N \sum_{z_i=1}^M \left(\theta_{z_i}(p, t) - \hat{\theta}_{z_i}(t) \right)^2 \right) \quad (17)$$

where N is the number of data points within the simulation time window, M is the number of soil depths considered for RZSM definition, $\theta_{z_i}(p, t)$ is the simulated soil moisture at depth node z_i at time t for the set of parameters (p) which are to be optimized and $\hat{\theta}_{z_i}(t)$ is observed soil moisture at depth node z_i at time t . For simplicity, all the depths are assigned equal weights. Alternatively, one has the flexibility to introduce varying weights for individual depth levels of soil moisture, thereby augmenting the complexity of the objective function.

2.3 | Numerical experiments and flow domain

Numerical experiments were conducted to estimate SHPs of DPM under different scenarios. Soil moisture data for these experiments was generated using forward modeling with SHPs for a macroporous soil domain as described by Šimůnek

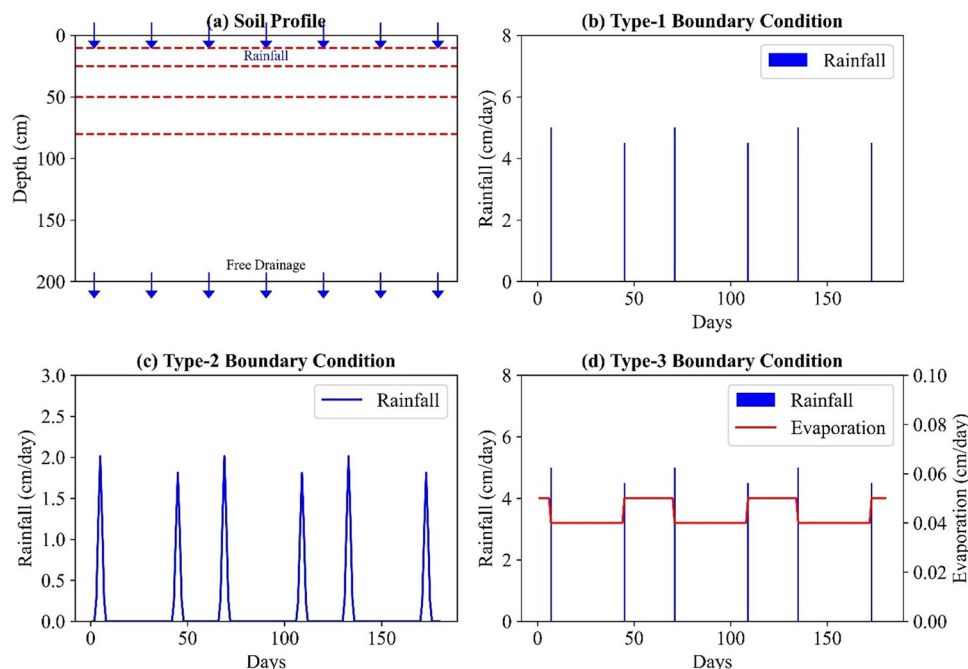


FIGURE 2 Schematic representation of model geometry and flow conditions: (a) model domain and soil moisture measurement grid locations (shown by dashed lines); (b) Type-1 upper boundary condition as recurring rainfall event; (c) Type-2 upper boundary condition as Gaussian distribution of rainfall event; (d) Type-3 upper boundary condition as rainfall with quasi-steady evaporation throughout the simulation window.

et al. (2003). A bare soil profile was simulated to isolate the effect of boundary conditions (Figure 2a). The depth of the soil was kept as 200 cm to avoid the influence of lower boundary.

To evaluate the uniqueness of the parameters identified by the optimization algorithm, three distinct boundary conditions were implemented (see Figure 2). Type-1 boundary condition attributes a recurring discrete rainfall event with a fixed pattern, while Type-2 boundary condition represents a gaussian distributed continuous rainfall in the system. The third type of boundary condition (Type-3) incorporates evaporation alongside the discrete recurring rainfall event (like conditions in a barren land) and representing real-world atmospheric boundary conditions. The following SHPs were utilized for the matrix, macropore and matrix–macropore interface: $\theta_{rm} = 0.105$, $\theta_{sm} = 0.50$, $\alpha_m = 0.005 \text{ cm}^{-1}$, $n_m = 1.5$, $l_m = 0.5$, $K_{sm} = 1.05 \text{ cm/day}$, $\theta_{rf} = 0.0$, $\theta_{sf} = 0.50$, $\alpha_f = 0.1 \text{ cm}^{-1}$, $n_f = 2.0$, $l_f = 0.5$, $K_{sf} = 2000.0 \text{ cm/day}$, $\omega_f = 0.05$, $\beta = 3$, $\gamma_w = 0.4$, $a = 1.0$, and saturated hydraulic conductivity of matrix–macropore interface, $K_{as} = 0.01 \text{ cm/day}$. As the saturated hydraulic conductivity of the matrix domain was very small, the interface mass transfer between matrix and macropore domain, the fluxes in matrix and the macropore domains were analyzed for varying K_{sm} values of 1, 5, 10, and 15 cm/day and for the boundary conditions shown in Figure 2. This was done to ensure that DPM is more suited for simulating flow in this soil compared to a single porosity model. The results (see Figure S1) indicated that lower K_{sm} led to greater,

more frequent and prolonged water flux exchange between the two domains.

SSM was defined as soil moisture at a depth of 10 cm. RZSM encapsulates values of soil moisture within the plant root zones. Since effective root zone for most crops extends up to 1 m (Fan et al., 2016), soil moisture at depths of 10, 25, 50, and 80 cm is representative of RZSM. These depths were assigned equal weights in the objective function and were deliberately selected because they correspond to actual observation depths in the field experiments, therefore ensuring consistency between field and numerical experiments. The DPM parameters were assumed to remain temporally constant.

Synthetic soil moisture data were generated using the DPM over a period of 180 days, corresponding to the duration of a typical cropping cycle, using the parameters taken from Šimůnek et al. (2003). For Type-1 and Type-2 conditions, the initial pressure head was set to -1000 cm , representing a relatively dry soil profile, to simulate infiltration driven by rainfall as the sole upper boundary condition. For Type-3 condition, initial pressure head was adjusted to -200 cm , representing wetter soil, to account for the combined effect of rainfall and evaporation, offering a more balanced and realistic infiltration process to effectively capture more complicated soil water movement dynamics. This intermediate moisture condition was adopted for a more balanced and realistic representation of the field conditions where infiltration is governed by evaporation in addition to rainfall, an upgrade over Type-1 and

Type-2 conditions in showing complexity of the soil moisture dynamics. The chosen pressure heads ensure that the soil model accurately reflects the given environmental conditions, promoting effective water movement into the soil.

To evaluate the sensitivity of inverse estimation to the assumed initial condition, a numerical experiment was performed under Type-2 boundary conditions using three initial pressure heads (−1000, −1500, and −2000 cm). The −1000 cm case corresponded to the initial condition of the observed soil moisture data, while the other two represented drier profiles. Parameters were estimated with both surface (SSM) and root zone (RZSM) soil moisture daily data, and the percentage error in the estimated parameters was computed (discussed in Section 4.1.4).

Soil moisture data at different spatial nodes and temporal resolutions, daily and weekly, were used as observed data for parameter identification. Although the synthetic soil moisture data from forward simulations were free from errors, real-world soil moisture data often include uncertainties or noise. To simulate this, 5% Gaussian noise was introduced into the synthetic data. This perturbation enabled the evaluation of the optimization algorithm's robustness in identifying parameters under noisy conditions.

2.4 | Modeling strategy

To simplify the optimization process, certain parameters were fixed based on values available in the literature (Simunek et al., 2003). Specifically, the tortuosity parameter l in Equation (11) was assumed to be 0.5 for both the domains and kept constant throughout the study. Similarly, the residual soil moisture in matrix (θ_{rm}) was fixed as 0.105, in macropores (θ_{rf}) as 0.0, and soil geometry parameters (β , a and γ_w) were fixed to their original values for rectangular slab type geometry (3, 1, and 0.4, respectively). The targeted parameters for optimization in the numerical experiment included the effective SHPs of matrix (θ_{sm} , α_m , n_m , and K_{sm}) and macropore (θ_{sf} , α_f , n_f , and K_{sf}) domains and volume fraction of macropore domain (ω_f). Since soil geometry and saturated hydraulic conductivity of matrix–macropore interface (K_{as}) are not generally available, K_{as} was also included as a parameter for optimization with macropore parameters. To investigate the effect of temporal resolution, data uncertainty, and parameter interdependence, different cases were formulated, and independent studies were performed as outlined in Table 1. In Case 1, four matrix parameters were optimized while keeping macropore, ω_f and K_{as} parameters fixed. Similarly, in Case 2, four macropore parameters along with ω_f were optimized while keeping the parameters of the other domain fixed. In Case 3, K_{as} was optimized alongside Case 2 parameters, allowing for an assessment of influence of K_{as} on other parameters. These cases were designed to

assess the performance of GA in estimating single-domain parameters.

Case 4 was a comprehensive case that aimed to optimize nine parameters (θ_{sm} , α_m , n_m , K_{sm} , θ_{sf} , α_f , n_f , K_{sf} , and ω_f) simultaneously. Finally, Case 5 was built upon Case 4 by including K_{as} , following the same rationale as in Case 3. Cases 1, 2, and 4 were optimized for all three boundary conditions described in Section 2.3 and eight types of datasets. Cases 3 and 5 were optimized under Type-3 boundary condition only and with four accurate datasets, as initial results indicated strong interdependence between K_{as} and ω_f (elaborated more in results section). Parameters were so optimized to ensure robust evaluation across varying conditions.

2.4.1 | Tuning of GA

The GA model was initially tuned using numerical experiments under Type-1 boundary condition optimizing only macropore domain parameters (θ_{sf} , α_f , n_f , and K_{sf}). This experiment focused on a scenario with the least number of observations, specifically a weekly SSM-based objective function. Various combinations of three population sizes and three mutation probabilities were tested, while the crossover probability remained constant, as listed in Table S1. Maximum fitness achieved by each combination plateaued after a few generations, indicating no further improvement (Figure S2).

The Nash–Sutcliffe efficiency (NSE) was calculated for the estimated parameters from each combination to evaluate their accuracy. Although the combination of a population size of eight and mutation probability of 20% resulted in the highest NSE, the combination with a population size of 12 and mutation probability of 40% was ultimately selected for analysis. This was mainly because the parameter estimates obtained from the latter combination were closer to the true values (Figure S3). Consequently, the tuned GA parameter set was adopted for DPM parameter optimization across different scenarios.

2.4.2 | Robustness of GA

The performance of GA across different parameter bounds: 25%, 40%, and 50% was analyzed. The objective was to determine whether the GA could effectively navigate both constrained and expansive search spaces.

From the datasets outlined in Table 1, Cases 1, 2, and 4 were analyzed for four distinct accurate data collection schemes. Type-2 boundary condition was chosen as macropore flow is more pronounced under continuous rainfall, significantly influencing the soil's water retention characteristics. The soil water retention curve (SWRC) was employed as a

TABLE 1 Description of cases considered for numerical experiments with three distinct boundary conditions and eight types of datasets. S: surface soil moisture, R: root zone soil moisture, D: data collected at daily interval, W: data collected at weekly interval, A: accurate soil moisture data, N: noisy (uncertain) soil moisture data.

Case	Optimized parameters	Fixed parameters	Boundary conditions	Type of dataset
Case 1	$\theta_{sm}, \alpha_m, n_m,$ and K_{sm}	$\theta_{sf}, \alpha_f, n_f, K_{sf}, \omega_f,$ and K_{as}	Type-1	SDA, SDN, SWA,
Case 2	$\theta_{sf}, \alpha_f, n_f, K_{sf},$ and ω_f	$\theta_{sm}, \alpha_m, n_m, K_{sm},$ and K_{as}	Type-2	SWN, RDA, RDN,
Case 3	$\theta_{sf}, \alpha_f, n_f, K_{sf}, \omega_f,$ and K_{as}	$\theta_{sm}, \alpha_m, n_m,$ and K_{sm}	Type-3	RWA, RWN
Case 4	$\theta_{sm}, \alpha_m, n_m, K_{sm}, \theta_{sf}, \alpha_f, n_f, K_{sf},$ and ω_f	K_{as}		
Case 5	$\theta_{sm}, \alpha_m, n_m, K_{sm}, \theta_{sf}, \alpha_f, n_f, K_{sf}, \omega_f,$ and K_{as}	–		

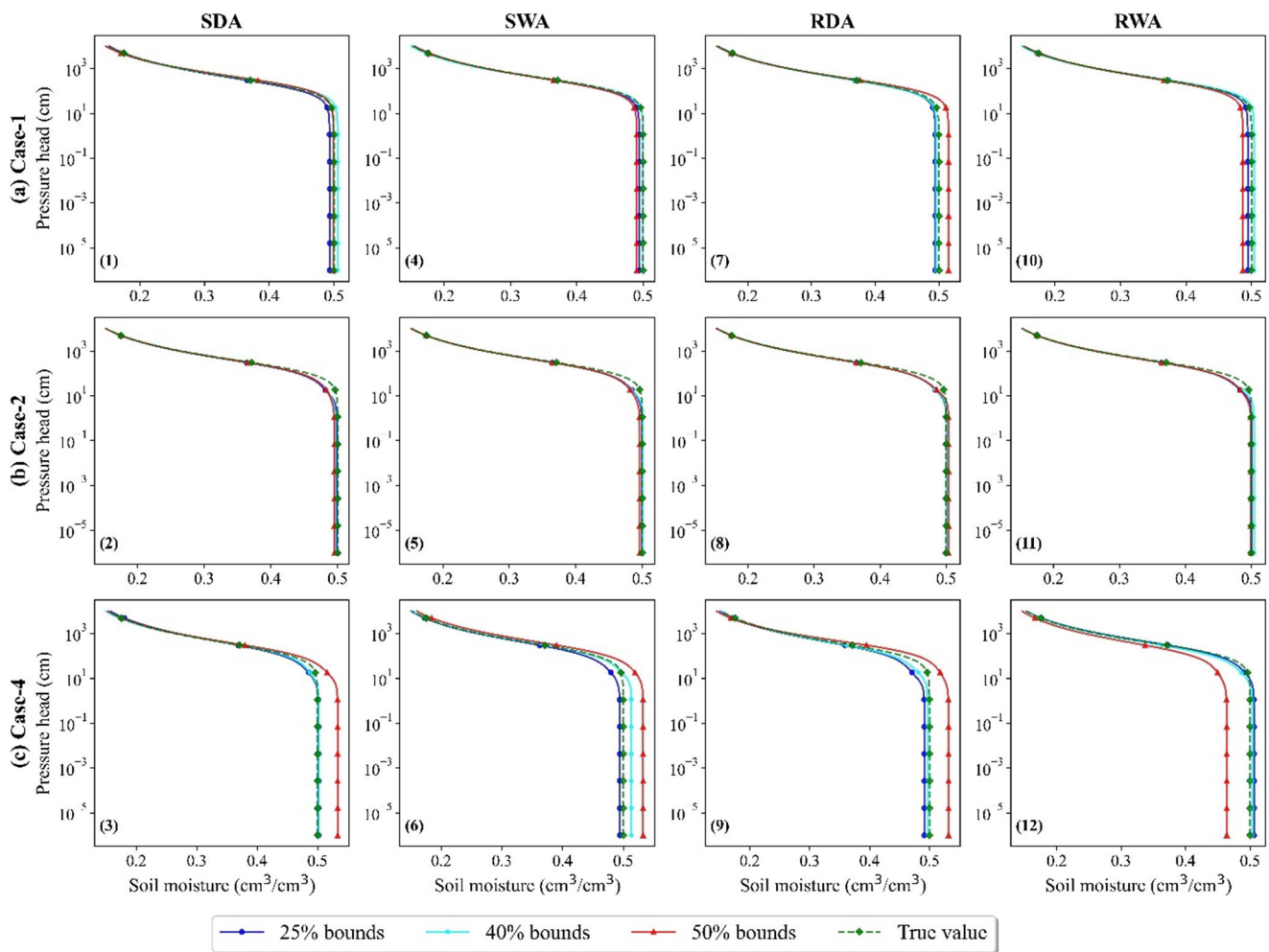


FIGURE 3 Soil water retention curve for different cases (Cases 1, 2, and 4) and different data collection schemes under Type-2 boundary condition with different parameter bounds (25%, 40%, and 50%). S: surface soil moisture, R: root zone soil moisture, D: data collected at daily interval, W: data collected at weekly interval, A: accurate soil moisture data.

benchmark to assess the GA's performance across different search spaces (Figure 3). Regardless of parameter bounds, Case 2 consistently yielded parameter estimates that were closest to the true values. This highlights GA's robustness in estimating macropore domain parameters, including ω_f .

Parameter bounds of 50% led to the most inconsistent results, particularly in Cases 1 and 4. These results frequently deviated from the true SWRC, often overpredicting or underpredicting soil moisture values. The inconsistency was attributed to the larger search space, which increased the likelihood of the

GA settling on physically implausible parameter values and becoming trapped in local minima.

Parameter estimates obtained with 40% bounds exhibited minor deviations. For instance, slight deviations were observed in the SWA scheme of Case 4. Despite these minor inconsistencies, the results were generally reliable compared to the 50% bound. SWRC estimates from 25% parameter bound showed excellent agreement with the true SWRC across all cases. This range allowed the GA to effectively converge on plausible parameter values while avoiding local minimum or unrealistic estimates. Based on the results, 25% parameter bounds were identified as the most robust and optimal for further analyses of θ_{sm} , α_m , n_m , K_{sm} , θ_{sf} , α_f , n_f , and K_{sf} . They provided the best balance between convergence speed, parameter plausibility, and consistency with the true SWRC. The parameter bounds for ω_f and K_{as} were kept as 0.02–0.15 and 0.01–0.10 cm/day, respectively, based on the ranges observed in literature (Gerke & van Genuchten, 1996; Pandey & Ojha, 2024) for efficient optimization of effective SHPs.

A detailed analysis of algorithm convergence and computational cost was performed for Type-1 boundary condition, as parameter identifiability is difficult for this boundary condition. To assess the convergence properties of the GA, the evolution of parameter gene values was studied throughout the simulation time (Figure S4) for daily RZSM using above average genes. The clustering of gene values around the true reference lines indicates stable convergence behavior. The convergence speed varied depending on the dimensionality of the optimization problem, as cases with fewer parameters (Cases 1 and 2) converged faster and required fewer evaluations. More complex cases (Cases 4 and 5) exhibited greater convergence time due to stronger parameter correlations (Table S2). Optimization using RZSM generally required higher runtime in simpler cases but achieved smaller runtime compared to SSM with more complex cases (Cases 3–5). Additionally, RZSM exhibited larger process memory requirement due to handling of multi-depth datasets.

2.5 | Efficacy evaluation of optimized parameters

To assess the adequacy of GA in optimizing model parameters and to evaluate the best-fit parameters across the selected study frameworks (as mentioned in Section 2.4), statistical analyses were performed. The uniqueness index (UI) of the optimized parameters under different boundary conditions was estimated using the following equation:

$$UI = (\hat{p}|_{95PCI+} - \hat{p}|_{95PCI-}) \times 100 \quad (18)$$

Here, $\hat{p}|_{95PCI+}$ is normalized value of upper bounds of estimated parameters, while $\hat{p}|_{95PCI-}$ is the normalized value of lower bounds of estimated parameters. A lower UI value indicates high uniqueness in the parameter estimates, signifying that the parameters are well-constrained and robust. Conversely, a higher UI value reflects poor uniqueness, suggesting that the parameter estimates may be uncertain or influenced by the boundary conditions.

In addition to this, statistical metrics like NSE coefficient, root mean square error (RMSE), mean absolute percentage error (MAPE), coefficient of determination (R^2), and mean bias error (MBE) were also calculated for comprehensive model evaluation.

3 | Field experiments and validation

3.1 | Study site

Field experiments were conducted in an experimental agricultural field (20 × 30 m) located at the Indian Institute of Technology Kanpur (26° 31' 0" N, 80° 13' 53" E; altitude 126 m above mean sea level) in the Upper Gangetic Plains of Uttar Pradesh, India. The plot size is representative of the small and scattered farm holdings typical of low-income farmers in the region. The mean annual precipitation is 801.5 mm, the mean annual temperature is 32.2°C, and the warmest and coolest months are May and January with a mean temperature of 41.3°C and 8.5°C, respectively (Panwar et al., 2018). The study site was divided into 24 subplots, each with an average area of approximately 11 m² (Figure 4a). The particle size distribution of soil at different depths was determined using laser diffraction technique. Based on the findings, sand fraction ranged between 12% and 15%, silt fraction ranged between 78% and 82%, and clay ranged between 6% and 8%. The soil was classified as silt loam. The irrigation volume was recorded after each application. The experiments focused on winter wheat as crop type, with observations conducted from January to April 2023 for a total of 102 days.

3.2 | Data collection

Irrigation water was applied to each subplot and measured using a water meter. The irrigation volume was determined based on the wheat-growing tradition of the area in consultation with the nearby farmers. For the present study, one subplot out of 24 was selected for experimentation and data collection. Plant growth parameters such as plant height, leaf area index (LAI), and root depth were measured at different crop growth stages. Rainfall was recorded at a 15-min interval using a tipping bucket rain gauge with a resolution of 0.2 mm and an accuracy of ±5% (Virtual Hydromet) installed

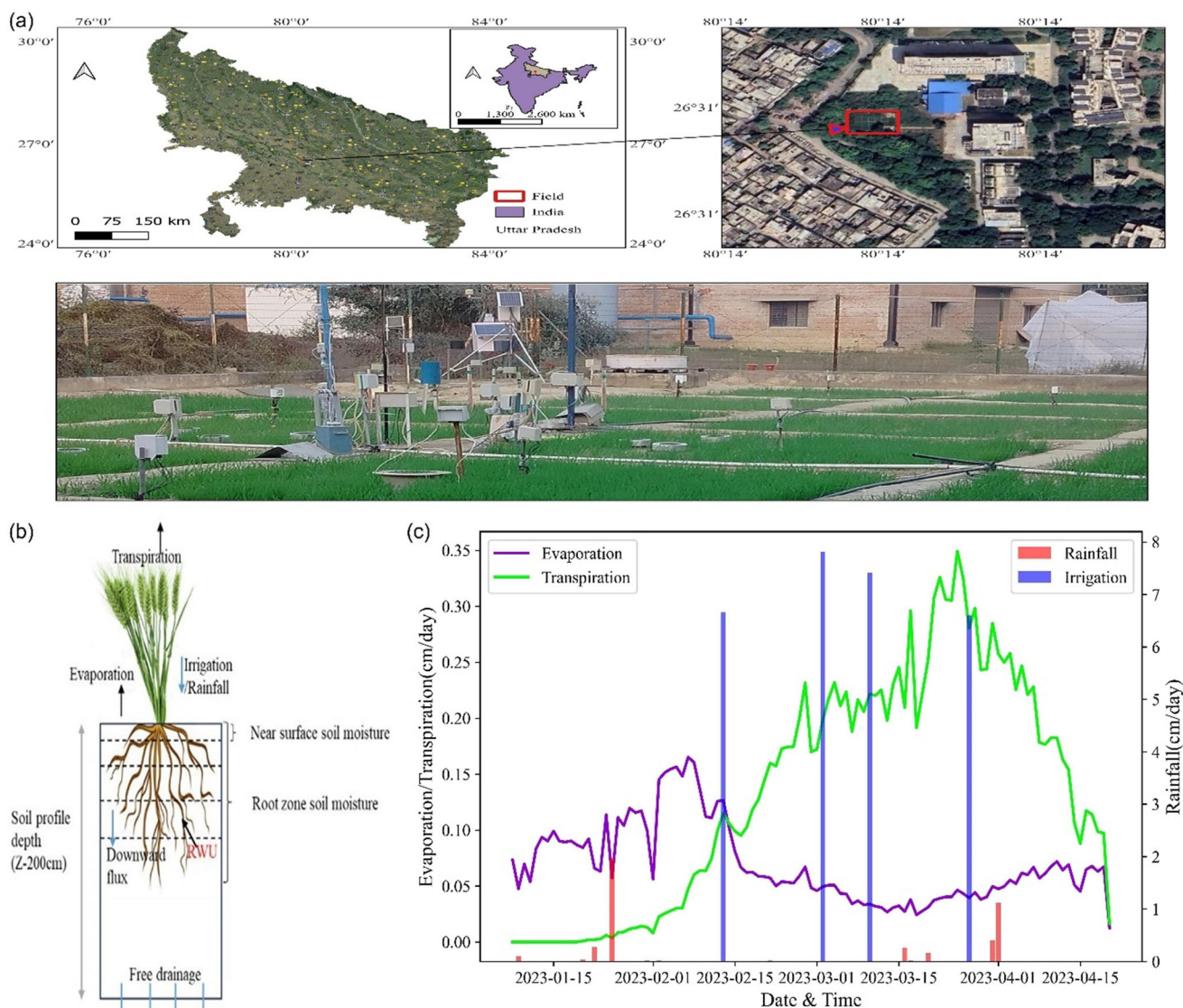


FIGURE 4 (a) Geographical location of the study area, (b) Schematic representation of model geometry and boundary conditions applied for simulations of field data, and (c) Time series of evaporation, transpiration, and rainfall and irrigation instances at the experimental site for the observation period.

at the field. Weather data such as wind speed, wind direction, relative humidity, solar radiation, temperature, etc. were measured using an Automated Weather Station (AWS) installed at the site at 15-min interval. The daily reference evapotranspiration (ET_0) was calculated using FAO 56 Penman–Montieth method (Allen et al., 1998) based on AWS data. ET_0 was multiplied by crop coefficient to obtain crop ET (ET_c). The daily ET_c was then partitioned into evaporation and transpiration using equations from Shankar et al. (2012) considering the crop's LAI. The field volumetric soil moisture observations were made using SM100 and SMEC300 soil moisture sensors (Spectrum Technologies) installed at the depths 10, 25, 50, and 80 cm. Both sensors have a reported accuracy of $\pm 3\%$ VWC. WatchDog 1000 series Micro Station was used to record the data with a 15-min time interval. Both SM100 (at

10, 25, and 50 cm) and SMEC300 (at 80 cm) soil moisture sensors were installed horizontally. Due to technical issues, the sensor placed at 25-cm depth ceased functioning after March 16, 2023. Figure 4b,c shows the domain setup and rainfall, evaporation, and transpiration variation during the study period.

For estimation of SHPs of DPM, field experiments were conducted using TDI (Soil Moisture Equipment Corp.) and double ring infiltrometer. Volumetric soil moisture measurements at the start and end of TDI experiment and after completion of double ring experiment were made using HydraGO (Royal Eijkelpkamp), which has measuring accuracy of $\pm 0.5\%$. The disc of 20 cm diameter was used for TDI experiments, and infiltration measurements were made at tensions of -12 , -9 , -6 , and -3 cm. The inner diameter of the double

ring infiltrometer was kept the same as that of TDI (20 cm), and a minimum water head of 10 cm was maintained until steady state infiltration rate was achieved. Ideally, both the matrix and macropore parameters can be estimated using the inverse module for DPM in HYDRUS-2D/3D (Simunek & van Genuchten, 1999). However, since we did not have access to the inverse module for DPM, a two-step approach was employed to estimate the matrix and macropore parameters. The SHPs (θ_{sm} , α_m , n_m , and K_{sm}) inversely estimated from the single porosity model in HYDRUS 2/3D from the TDI measurements were assumed to characterize matrix domain. The macropore parameters of the DPM (θ_{sf} , α_f , n_f , K_{sf} , and ω_f) were estimated using cumulative infiltration data from double ring infiltrometer experiment using HYDRUS-1D, fixing the matrix domain parameters obtained in previous step. Repeated simulations were performed with varying parameter sets to achieve the best fit between simulated and observed soil moisture profiles. The parameter set yielding the most favorable results was selected. It is to be noted that HYDRUS-1D and HYDRUS-2D/3D use Levenberg–Marquardt algorithm for inverse parameter estimation.

3.3 | Experimental validation of parameters

The daily and weekly interval SSM and RZSM were used to inversely estimate the parameters. This provided us with four types of datasets. The soil matrix and macropore parameters obtained in the previous section were used to define the search bounds ($\theta_{rm} = 0.05$, $\theta_{sm} = 0.29$, $\alpha_m = 0.001$, $n_m = 1.7$, $l_m = 0.5$, $K_{sm} = 4$ cm/day, $\theta_{rf} = 0.0$, $\theta_{sf} = 0.35$, $\alpha_f = 0.01$, $n_f = 2.1$, $l_f = 0.5$, $K_{sf} = 450$ cm/day and $\omega_f = 0.05$). The values of remaining interface mass transfer parameters (β , a , γ_w , and K_{as}) were taken from Gerke and van Genuchten (1996) for rectangular slab ($\beta = 3$, $\gamma_w = 0.4$, $a = 1.0$, and $K_{as} = 0.01$ cm/day). The depth of soil profile was kept at 200 cm to avoid boundary effects. The simulation was done for wheat crop cover, and root water uptake model of Ojha and Rai (1996) was used. Initial condition in terms of pressure head for both fracture and matrix domains was provided as -2000 cm to match the conditions observed in the field near the soil surface at the time of sowing. Four objective functions comprising SSM and RZSM at daily and weekly intervals were formulated. The parameter combinations formulated for inverse estimation were same as those described in Table 1. The system was subjected to atmospheric boundary conditions observed in the field with wheat crop cover (see Figure 4c).

4 | RESULTS AND DISCUSSION

This section contains results and discussion related to parameter estimation of numerical experiments along with suitability

of these parameters in generating soil moisture, followed by results for in situ validation experiment.

4.1 | Numerical case study

4.1.1 | Estimation of matrix parameters

Figure S5 shows the average values of van Genuchten-based soil matrix parameters and the 95% confidence interval (95PCI) for Case 1 under various datasets and boundary conditions (the values are listed in Table S3). Further, the uniqueness of estimated parameters is shown in Figure 5.

Here, θ_{sm} exhibited high identifiability across various boundary conditions, with a relative error of $<3\%$. Under Type-2 boundary condition, challenges were noted in precisely estimating θ_{sm} likely due to gradual saturation of the system. Inverse of air entry pressure, α_m , which is the representative of the initial portion of SWRC, was estimated accurately (relative error being $<8\%$ across all the boundary conditions and data collection schemes), with easiest identification during Type-2 boundary condition, suggesting easier prediction with more dynamic and continuous rainfall. The K_{sm} was the least accurately predicted parameter, showing high error (Figure S5). The difficulty in estimating K_{sm} stems from its exceptionally small true value, which poses challenges for the GA optimization process.

In the context of temporal scale, daily RZSM data yielded more accurate estimates of effective SHPs compared to lower temporal resolution data. As evaporation is a surface phenomenon, the compromise in temporal scale of SSM may invert the parameter identifiability trend, which can be seen in Figure 5. The results for soil moisture with noise are contrary to those obtained with accurate soil moisture data. Specifically, θ_{sm} and n_m were estimated with less accuracy compared to precise soil moisture data. Conversely, parameters K_{sm} and α_m showed improved accuracy in their estimation. Nevertheless, it is evident that, overall, α_m and K_{sm} were consistently underestimated when uncertain soil moisture data was used, except in the case of weekly soil moisture.

In general, the parameters estimated using the accurate soil moisture data displayed higher uniqueness. Here, θ_{sm} , n_m , and α_m consistently showed a high degree of uniqueness across various datasets. The uniqueness of K_{sm} varied with boundary conditions and data type. Despite high accuracy in parameter estimates derived from perturbed soil moisture data, the reduced precision indicates less unique solutions to the optimization problem.

4.1.2 | Estimation of macropore parameters

In Case 2, the effective SHPs of macropore domain (θ_{sf} , α_f , n_f , and K_{sf}) along with volume fraction of macropore (ω_f)

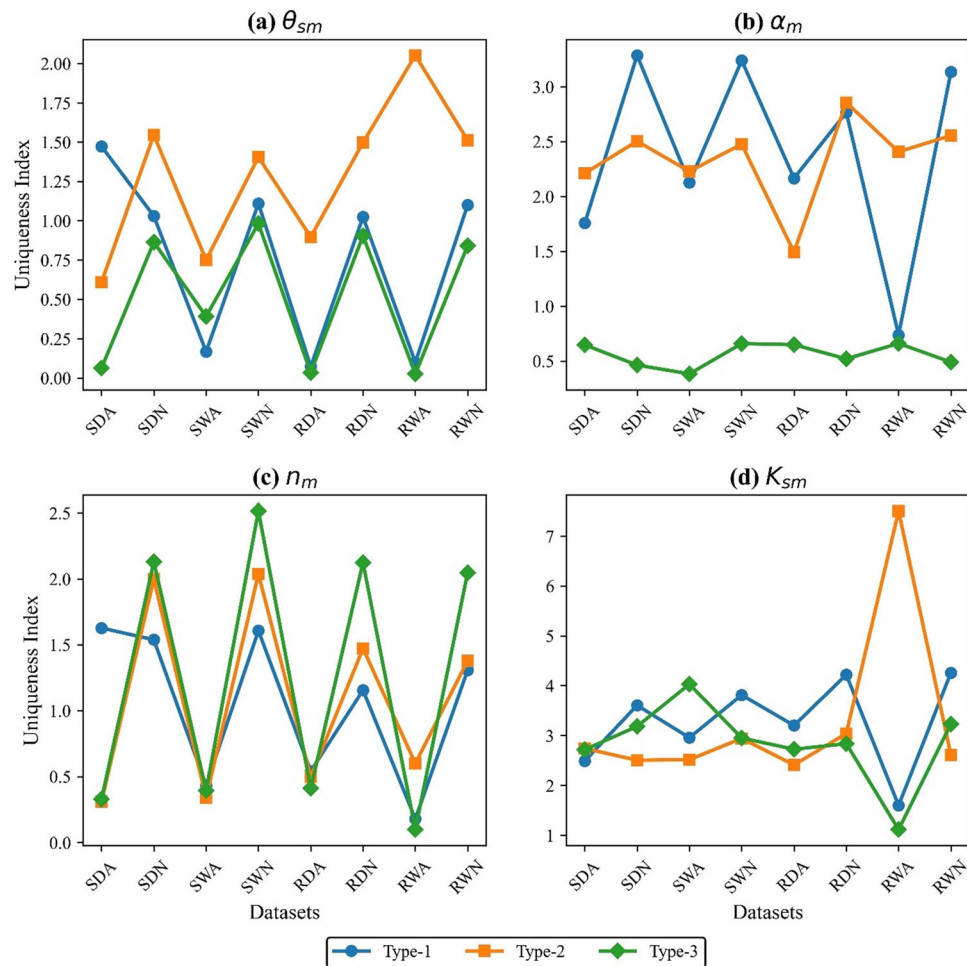


FIGURE 5 Uniqueness index of optimized parameters for different datasets with all three boundary conditions when matrix parameters were optimized (Case 1). S: surface soil moisture, R: root zone soil moisture, D: data collected at daily interval, W: data collected at weekly interval, A: accurate soil moisture data, N: noisy (uncertain) soil moisture data.

were estimated while fixing the matrix parameters (Table S4). In this case, θ_{sf} was easily identified using accurate data with not >6% error, particularly with daily RZSM data (Figure S6). The best estimates for θ_{sf} , α_f , n_f , and ω_f were achieved with RZSM at daily interval, except for K_{sf} , which was better identified with SSM data. The estimated parameter set was sensitive to the variation in ω_f . As ω_f approached its true value, the error in other parameters reduced. However, with uncertain soil moisture data, accurate estimation of ω_f becomes challenging leading to disturbances in other parameters, especially in θ_{sf} . Surprisingly, the SHPs of the macropore domain estimated using uncertain soil moisture data demonstrated higher uniqueness in some cases compared to accurate soil moisture data. In general, the parameters estimated using accurate soil moisture data showed high uniqueness, while the perturbed soil moisture data led to poorer uniqueness. Among the parameters, n_f exhibited a high degree of uniqueness across all conditions, while K_{sf} and α_f showed moderate variations in their uniqueness.

In Case 3, optimization was performed only with the accurate dataset under Type-3 boundary condition. The effective SHPs of the macropore domain (θ_{sf} , α_f , n_f , and K_{sf}) along with the macropore volume fraction (ω_f) and the saturated hydraulic conductivity of the interface (K_{as}) were estimated while maintaining fixed matrix parameters. The parameter K_{as} was well-identified as 0.015 cm/day using RZSM data at daily resolution. However, the ω_f estimation exhibited error under this temporal resolution. It is likely because of the cross-correlation between these two parameters. Notably, K_{as} showed highest error when optimized with daily SSM data, whereas this dataset showed minimal residual error in ω_f to 0.17%.

The identifiability of θ_{sf} was compromised when K_{as} was co-optimized with macropore parameters increasing its estimation error from 6% in Case 2 to 13% in Case 3 (Figure 6). While optimization of α_f improved marginally (1%–2%) across various datasets (SDA, SWA, RDA, and RWA) in Case 3, the estimation of n_f deteriorated, with errors escalating to 6%–14%. Similarly, K_{sf} exhibited increased uncertainty,

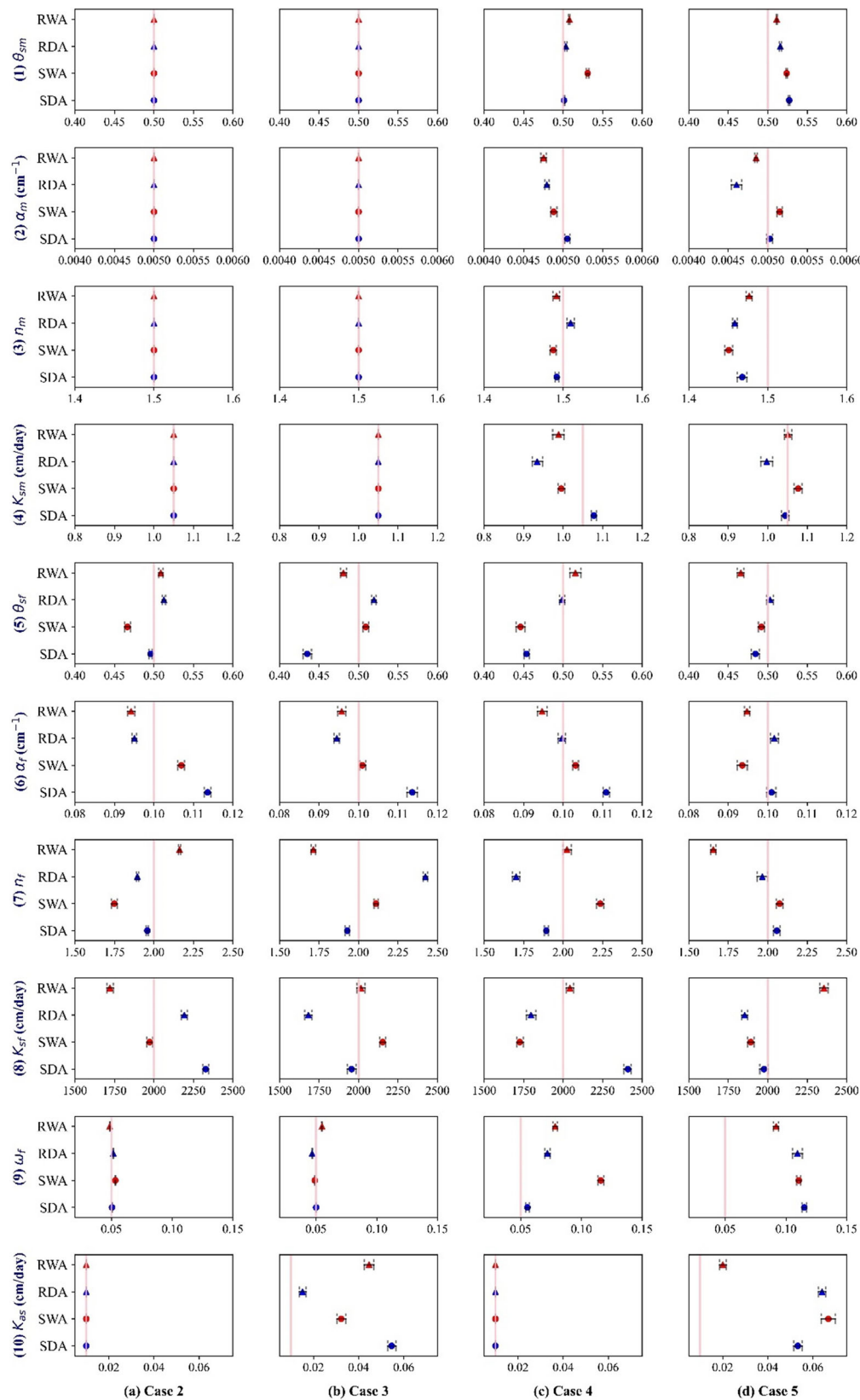


FIGURE 6 Estimates of effective soil hydraulic parameters of matrix and macropore domain and their 95PCI for different parameter combinations under Type-3 boundary condition. Circular marker represents parameters optimized using SSM, while triangular marker shows root zone soil moisture (RZSM)-based optimized parameter. The red vertical line denotes the true value. S: surface soil moisture, R: root zone soil moisture, D: data collected at daily interval, W: data collected at weekly interval, A: accurate soil moisture data, N: noisy (uncertain) soil moisture data.

with error rising from 9.6% in Case 2 to 15% in Case 3 with RZSM at daily scale. Overall, while estimating K_{as} in Case 3 yielded minor improvements in α_f , it led to systematic decline in accuracy across multiple SHPs, underscoring the trade-offs in parameter identifiability under the chosen optimization strategy.

4.1.3 | Estimation of matrix and macropore parameters

In Case 4, nine parameters, four of soil matrix and five of the macropore domain, were estimated together. Table S5 and Table S6 present the results of matrix and macropore parameters, respectively. In this case, identification of matrix parameters was much easier as compared to that of macropore hydraulic parameters. The θ_{sm} and n_m were estimated with highest accuracy among all parameters (Figure 7). Conversely, the error in their macropore counterpart (θ_{sf} , α_f , and n_f) increased while using accurate data. Inaccurate estimation of ω_f may introduce error in other parameters of macropore domain. Despite these challenges, macropore parameters were still estimated with a relative error of <14%.

The scale parameters of macropore domain were highly sensitive to variation in ω_f . The K_{sm} and K_{sf} showed the least precise estimates out of all the parameters. The macropore domain parameters were estimated to have lower relative error and higher uniqueness (Figure 7) when computed using uncertain soil moisture data across different boundary conditions. Out of all the identifiers used in this study, RZSM at daily time interval proved to be most effective, particularly under highly dynamic and variable boundary conditions. As the dimensionality of the optimization framework increases, the GA search space yields a better fit but often lacks physical realism, as parameters may deviate from their true values. This highlights the strong correlation among parameters. The simultaneous estimation of large number of parameters introduces correlation between them, which cannot be ignored. In general, the model performed well across all temporal scales, demonstrating robustness in parameter estimation.

Like Case 3, parameters in Case 5 were optimized under Type-3 boundary condition with accurate datasets. Introduction of K_{as} as an optimization parameter (in Case 5) posed significant optimization challenge, with estimated value of K_{as} ranging from five to seven times (0.05–0.067) the true value (0.01 cm/day), highlighting the difficulty of constraining this parameter even when working with synthetic soil moisture data and known parameter values. The residual percentage error in ω_f increased significantly, ranging from 82% to 130% across different datasets and estimation schemes. Estimation of K_{sf} and n_f showed improvement. However, not much difference was observed in estimates of other parameters. Fixing interface parameters restricts the optimization

potential, whereas including them in the estimation process improves the accuracy of all macropore hydraulic parameters (θ_{sf} , α_f , n_f , and K_{sf}). Overall, SSM and RZSM data at daily scale yielded superior parameter estimates in comparison to weekly observations, as illustrated in the forest plot (Figure 6).

4.1.4 | Suitability of estimated parameters in generating soil moisture

The optimized parameters of different cases were used to simulate soil moisture depletion curves under different boundary conditions. The soil moisture was significantly underpredicted across all depths and for all three boundary conditions when using parameter set obtained from Case 1 (Figure S7). This is attributed to the inaccurate estimation of α_m . Evidently, even though the matrix parameters were estimated with high uniqueness, they might not give a clear picture of the underlying dynamics when optimized in isolation from macropore parameters. The predicted soil moisture curves improved significantly when simulated using Case 2 parameter combination (Figure S8), except under Type-1 boundary condition, which remained challenging due to the discrete and instantaneous nature of rainfall events. The results for Case 4 are shown in Figure 8 for accurate data. A good match was observed between the simulated and observed soil moisture across all depths.

The NSE, RMSE, R^2 , and MBE values for different cases across three distinct boundary conditions and 8 types of datasets were estimated for the four depths, and the results are shown for Cases 1, 2, and 4 (Figure 9; Figure S9). NSE values >0.5 were considered as a good fit, and values near to 1 were considered as the best fit.

When only matrix parameters were optimized (Case 1), under Type-1 and Type-3 boundary conditions, high variability was observed in NSE, implying that model performance is less reliable. The predicted soil moisture was consistently underestimated across all the depths, regardless of the data source (SSM or RZSM). The model was not able to effectively capture the inherent properties of soil matrix in case of discrete instantaneous rainfall. The variability seemed to reduce slightly under Type-2 boundary condition, reflecting better model performance due to continuous and prolonged rainfall. On the other hand, during Case 2 analysis, higher and more consistent values of NSE were obtained. Despite increased variability in Type-3, NSE values were still relatively high given the complexity of boundary conditions, demonstrating the robustness of the model even under more complex conditions. The consistency of accurate data was found to be better than noisy data, and similar trends of soil moisture were observed for both surface and root zone data.

Including all parameters in the estimation did not hamper the model performance in Case 4, with high consistency

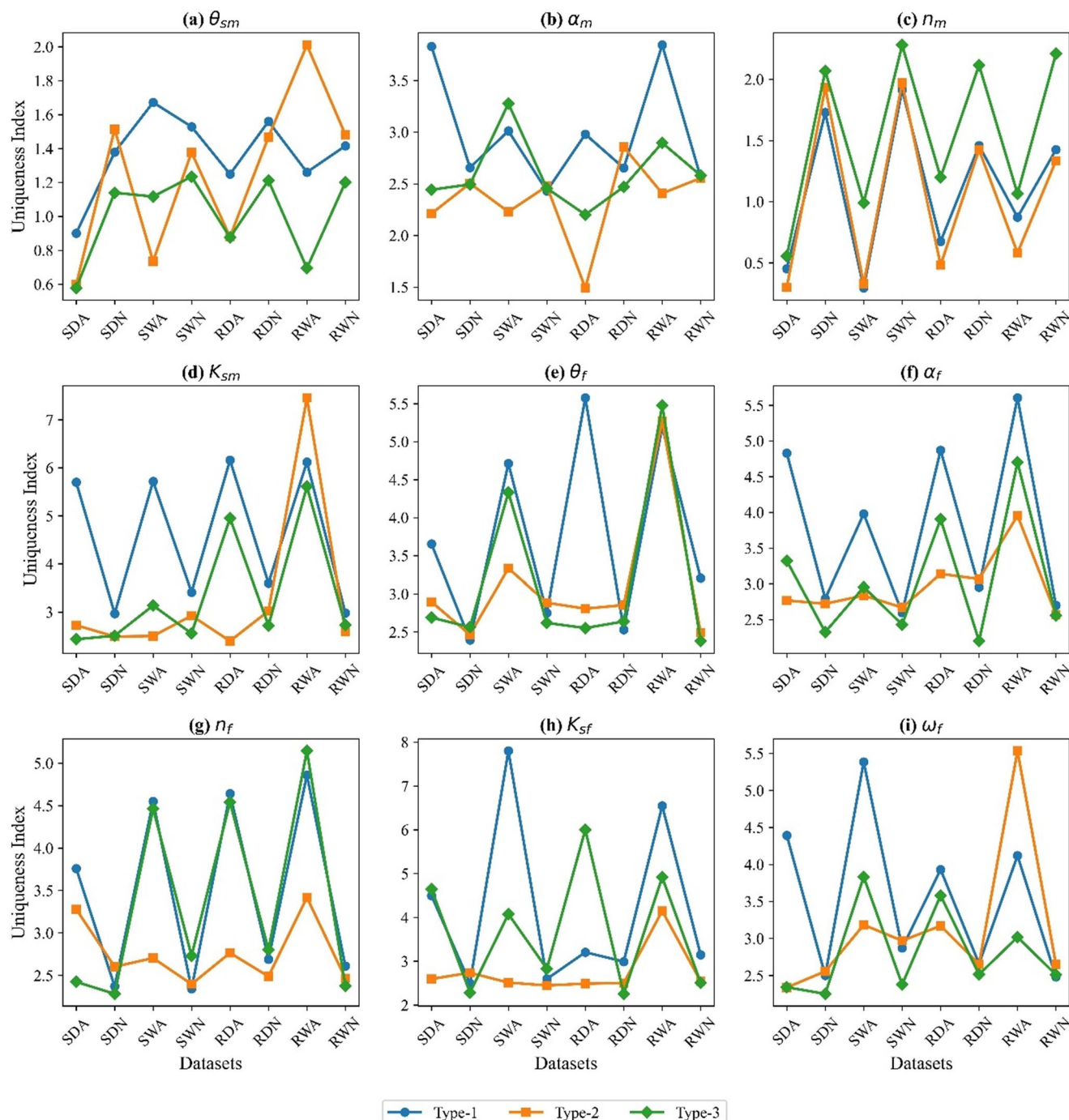


FIGURE 7 Uniqueness index of optimized parameters for different soil moisture datasets under all three boundary conditions with Case 4 parameter combination. S: surface soil moisture, R: root zone soil moisture, D: data collected at daily interval, W: data collected at weekly interval, A: accurate soil moisture data, N: noisy (uncertain) soil moisture data.

in NSE values, close to 1, observed across all the boundary conditions. The best model performance was observed in Type-2 boundary condition, highlighting the effectiveness of GA in estimating all parameters under continuous rainfall event. Occasional outliers were observed in Type-1 condition and a slight increase in variability was encountered in Type-3 condition. Nevertheless, the model robustly estimated the parameters even under complex and near-real field scenarios.

RMSE values were generally low (≤ 0.07) across all cases, depths, and boundary conditions, with maximum value observed for Case 1 at each depth and three boundary conditions (Figure S9). A similar trend was observed in MBE as well as R^2 where Case 1 showed maximum bias and highly negative R^2 across all depth and with all three types of boundary conditions. This showed that estimating matrix parameters alone would be insufficient and reduce predictive

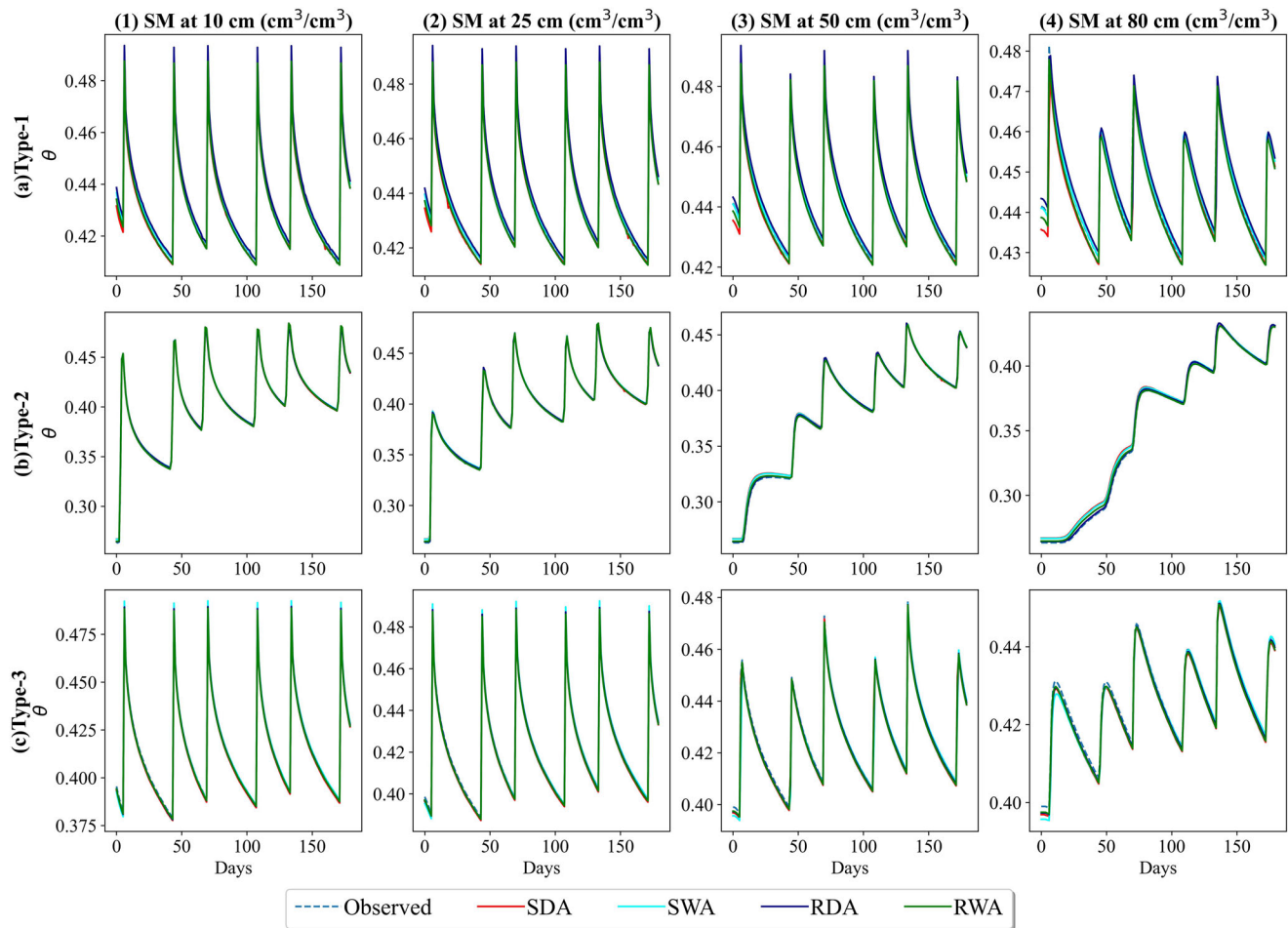


FIGURE 8 Soil moisture variation over simulation period in numerical experiment across different depth when 9 soil hydraulic parameters (SHPs) were simultaneously estimated (Case 4) using accurate soil moisture data with (a) Type-1 boundary condition, (b) Type-2 boundary condition, and (c) Type-3 boundary condition. S: surface soil moisture, R: root zone soil moisture, D: data collected at daily interval, W: data collected at weekly interval, A: accurate soil moisture data. SM: soil moisture

strength of the algorithm. On the other hand, Cases 2 and 4 showed nearly zero bias and RMSE, and R^2 almost 1, suggesting the significant influence of macropores as well as joint macropore-matrix interactions on soil moisture. Notably, daily RZSM provided better results, especially at deeper depth under all types of boundary condition and datasets. The consistently small bias suggested that GA-based inversion did not introduce over- or under predictions in soil moisture.

During the test with variable initial conditions (−1000, −1500, and −2000 cm), the errors in estimated parameters grew with increasing deviation from the true initial condition, especially for macropore properties (θ_{sf} , α_f , and K_{sf}), with ω_f showing the largest error under daily SSM (Figure S10d). This error propagation was also reflected in simulated soil moisture, with greater error observed with depth and for initial condition farther from the true one (Figure S11). These findings corroborate earlier reports that inaccurate initial conditions bias parameter identification and

degrade model performance (Bandai & Ghezzehei, 2022; Yu et al., 2019). This work extends that evidence to DPM, underscoring the need for careful identification of initial conditions in inverse modeling for parameter estimation.

4.2 | In situ experimental validation

The proposed GA-based framework was used to optimize parameters for cases described in Table 1 using in situ soil moisture data. During estimation of matrix parameters (Case 1), θ_{sm} , α_m , and n_m were uniquely identified for surface and root zone data across all temporal scales (maximum UI being 1.42%, 3.09%, and 3.01% for θ_{sm} , α_m , and n_m , respectively) (Figure 10a). The identification of K_{sm} using in situ data was found to be more challenging compared to the other parameters, with the lowest uniqueness observed across all data schemes (maximum value of UI reaching 3.97% for surface weekly soil moisture). This observation aligns with the

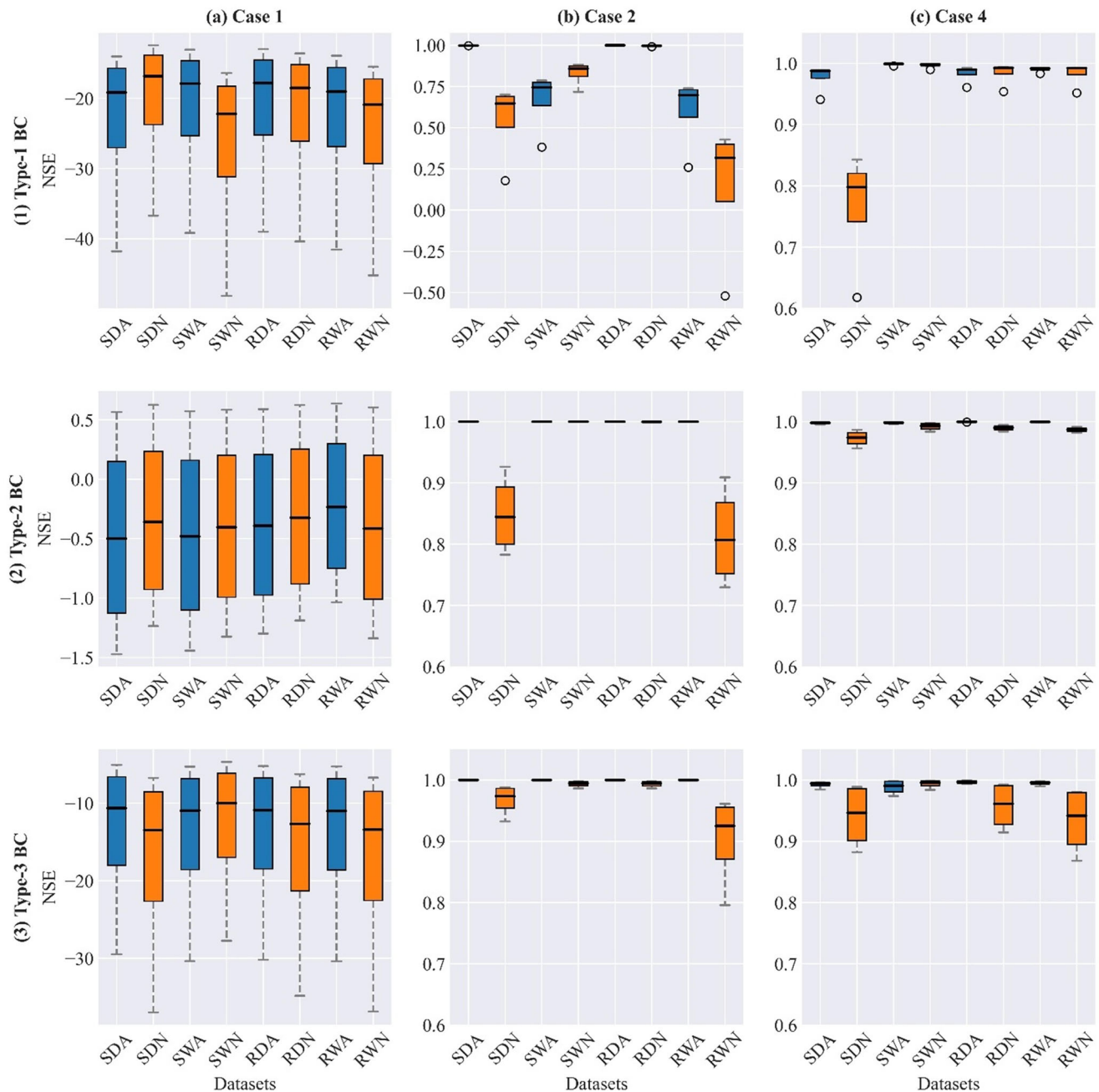


FIGURE 9 Model-predicted NSE values of soil moisture with optimized parameters across all 8 datasets under all three boundary conditions for (a) Case 1, (b) Case 2, and (c) Case 4; S: surface soil moisture, R: root zone soil moisture, D: data collected at daily interval, W: data collected at weekly interval, A: accurate soil moisture data, N: noisy (uncertain) soil moisture data, NSE: Nash–Sutcliffe efficiency. Note that the scale of y-axis is different across subplots.

observations from numerical experiments using accurate soil moisture data (Section 4.1.1).

In Case 2, ω_f was estimated with high uniqueness (UI < 0.97%). As discussed in Section 4.1.2, accurate estimation of ω_f would assist in better estimation of macropore parameters, which is reflected in the high uniqueness of the remaining four parameters (UI < 3.09%) (Figure 10b). This also shows that the proposed model can estimate macropore and ω_f parameter robustly even under complex real-time field conditions. The performance of the proposed model did not

decline while using RZSM data, achieving similar uniqueness with surface data. Case 4 parameter combination with SSM at weekly scale consistently estimated parameters with low uniqueness (Figure 10c). Nevertheless, all the parameters were uniquely estimated using other datasets with UI < 2.8%. Additionally, RZSM observations consistently yielded more accurate parameter estimates, which align with observations from numerical experiments.

Similar to numerical experiments, for the field, where precise soil geometry is unavailable, interface hydraulic

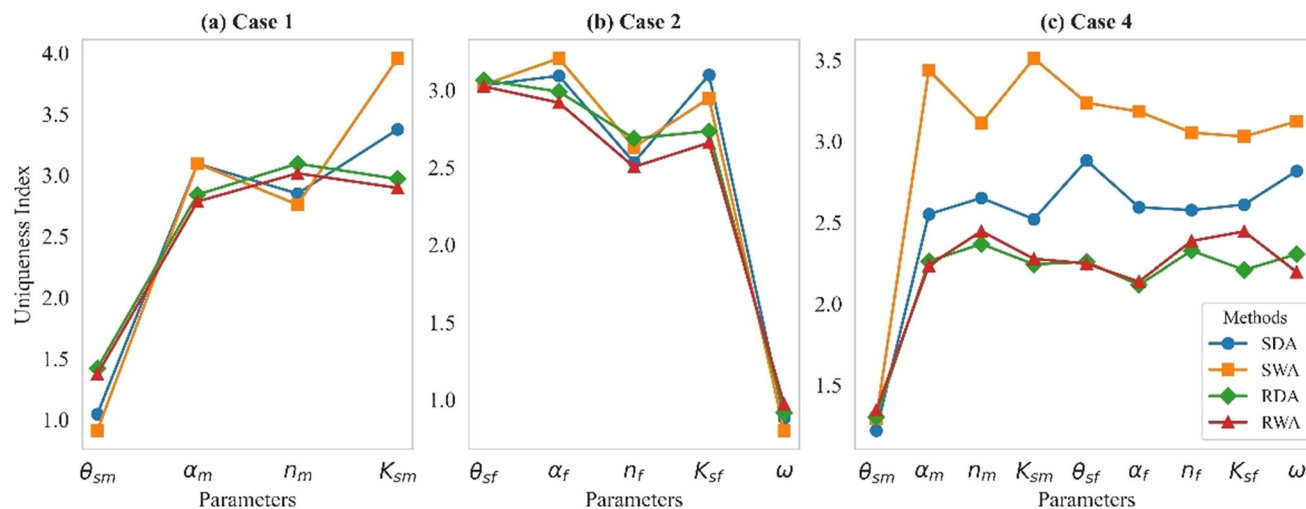


FIGURE 10 Uniqueness index of optimized field parameters using different datasets for (a) Case 1, (b) Case 2, and (c) Case 3. S: surface soil moisture, R: root zone soil moisture, D: data collected at daily interval, W: data collected at weekly interval, A: accurate soil moisture data (here, observed data).

conductivity (K_{as}) was estimated. Parameter optimization was performed using both SSM and RZSM data at daily and weekly temporal scales. The results indicate that the macropore connectivity parameter (ω_f) is consistently well-constrained, even without explicitly optimizing K_{as} (Figure 11). In both Cases 2 and 3, the estimated values of ω_f remain consistent, except when using RZSM data at a daily scale, which yields estimates closer to the actual value with reduced percentage error.

Optimizing K_{as} alongside other macropore parameters (Cases 2 and 3) significantly reduced the relative error in estimated θ_{sf} and n_f while targeting RZSM at daily scale. The relative error in θ_{sf} reduced from 8.16% in Case 2 to 4.7% in Case 3. Similarly, n_f showed a 9% improvement, reducing the relative error to 2.34%. No significant improvement was observed in estimation of K_{sf} with relative error averaging around 4.54% and 6.55% in Cases 2 and 3, respectively, across all datasets. In case of α_f , the estimate seemed to worsen, increasing the relative error by 6.77%.

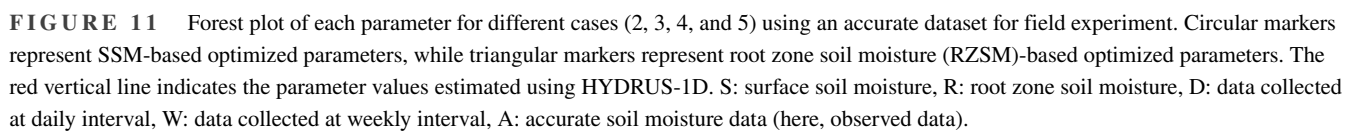
Parameter estimates in Cases 4 and 5 showed improvement over Cases 2 and 3; that is, when matrix domain parameters were optimized together with macropore domain. However, no significant reduction in relative error of estimation parameters was observed when K_{as} was included in estimation along with other nine parameters. For example, the error in estimated θ_{sf} ranged between 0.05% and 1.63% with different datasets for Case 5 as opposed to 1.6%–3.5% in Case 4. Similarly, the relative error in n_f lied between 0.13% and 2.5%, Case 5 whereas it ranged between 1.4% and 2.66% in Case 4. The K_{as} estimated with RZSM at daily time scale (0.0513 cm/day) was found to be significantly higher than the assumed value (0.01 cm/day) but still falling within literature-reported range of K_{as} (0.01–0.1 cm/day) by Gerke and van Genuchten (1996). Notably, optimizing K_{as} hin-

dered the identification of ω_f , with difference in the relative error between Cases 4 and 5 reaching as high as 10.6%. This implies that the model can predict effective SHPs well in the absence of K_{as} and soil geometry information as long as ω_f is accurately provided.

A similar trend is reflected in the soil moisture obtained using the parameters from these four cases. The soil moisture simulation, using the optimized parameters, was conducted with a spin-up period until the percentage error between observed and simulated soil moisture fell below 1%. This time step signifies the attainment of steady-state conditions, where the model effectively captures the dynamics of the field environment, and the influence of inaccurate initial conditions is minimized. In our case, this condition was achieved on the 38th day of simulation. Therefore, the comparison between observed and simulated soil moisture is presented for the period from the 38th day until the end of the crop season. The 25-cm depth was excluded from this analysis, as only 28 days of observed soil moisture data remained after the spin-up period (Figure S12).

Cases 4 and 5 resulted in improvement in soil moisture simulations with reduced relative error as compared to Cases 2 and 3 (Figure 12). Average percentage relative error in predicted soil moisture at 10, 50, and 80 cm ranged between 2.67% and 17.34%, 5.10% and 10.94%, and 3.91% and 9.69%, respectively. Cases 2 and 3 exhibited greater relative error in simulated soil moisture (average maximum = 26.68% for SSM at weekly scale in Case 3). Minimum average % relative error in soil moisture was observed at 10 cm depth with Case 4 at daily scale, while maximum was observed with Case 3 at 10 cm using daily data.

The computational cost analysis for field experiment revealed notable performance disparities between the SSM and RZSM approaches for parameter estimation (Figure S13).



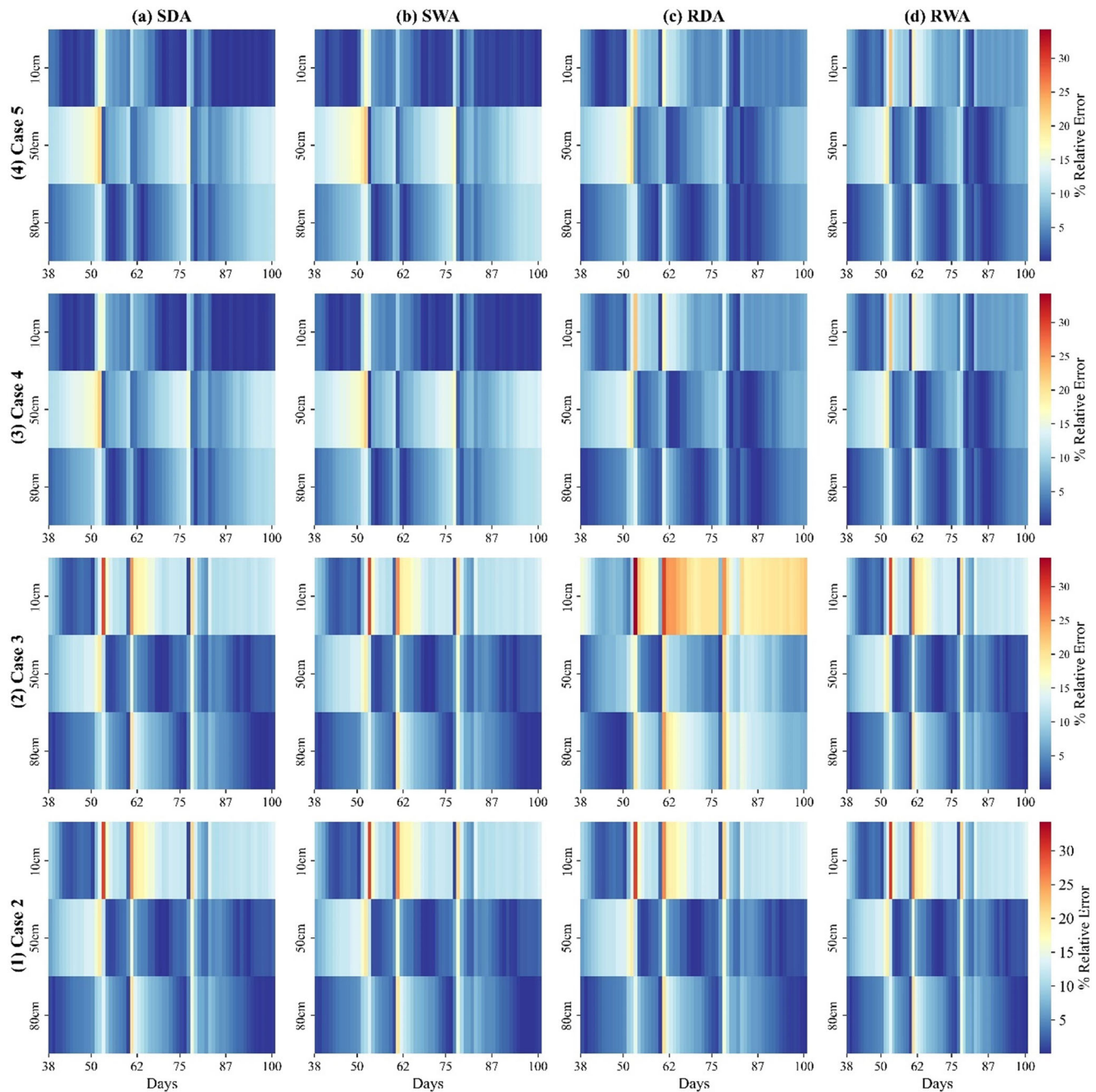


FIGURE 12 Percentage relative error in observed and simulated soil moisture obtained for different combinations of optimized parameters (Cases 2, 3, 4, and 5) with various datasets S: surface soil moisture, R: root zone soil moisture, D: data collected at daily interval, W: data collected at weekly interval, A: accurate soil moisture data (here, observed data).

The RZSM method converges more effectively toward the true values of SHPs by leveraging enhanced genetic information derived from RZSM moisture data, thereby reducing the time required for individual simulations and consequently lowering the overall runtime. In contrast, the SSM approach suffers from GA inefficiencies, introducing suboptimal genes that necessitate extended execution times, resulting in a higher total simulation duration. Nevertheless, the improved accuracy of RZSM is accompanied by elevated data processing

requirements, leading to increased process memory usage (Figure S13d).

5 | IMPLICATIONS OF USING DPM

In agricultural fields, both water drainage (outflow) and the redistribution of water within the soil profile take place during a cropping season (Raimbault et al., 2021). Presence

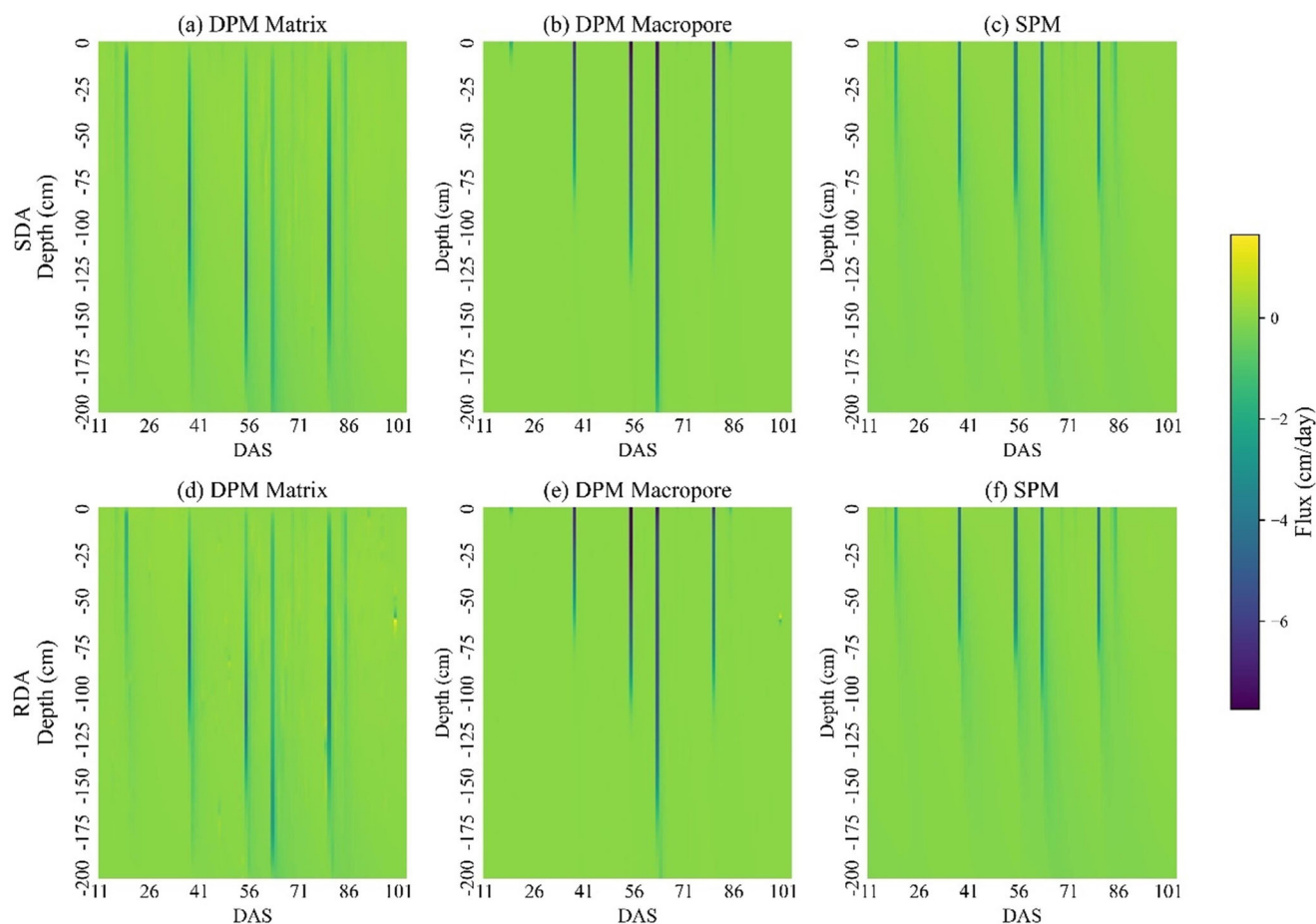


FIGURE 13 Flux from matrix and macropore domain along the soil profile from 10th day after sowing to end of crop period (102 days). (a) Flux in soil matrix of DPM, (b) flux in macropore of DPM, (c) flux in matrix of single porosity system. S: surface soil moisture, R: root zone soil moisture, D: data collected at daily interval, A: accurate soil moisture data (here, Observed data), DAS, days after sowing; DPM, dual permeability model; SPM, single porosity model.

of macropores in agricultural fields can aggravate this phenomenon. In a soil system with macropores, the soil matrix receives flux at intermediate depths from the highly permeable macropore domain, leading to an increase in flux and, consequently, elevated soil moisture levels at these depths. Macropores also facilitate the redistribution of soil moisture to deeper layers. A single porosity-based model cannot capture the transfer of fluxes between macropore and matrix and the overall redistribution of soil moisture.

For the field site, the effects of using SPM in comparison to DPM were studied. Matrix parameters of SPM were optimized using the GA-based framework. Figure 13 shows the fluxes in the matrix and macropore domains in a dual permeable system, and matrix flux in a single-permeability system for the field site. In the SPM framework, the soil matrix showed a uniform wetting pattern, whereas in the DPM framework, the wetting front was non-sequential. This non-uniform wetting behavior is a common observation in agricultural fields (Nimmo, 2010, 2012; Nimmo & Mitchell, 2013). The DPM model was also able to better predict soil

moisture when compared to SPM, especially at deeper depths. The MAPE in soil moisture for DPM is 3.92% at 80 cm depth as opposed to 8.68% for SPM, thereby highlighting the advantage of employing dual permeable models in agricultural fields (Figure S12).

To understand the practical implications of using DPM against SPM in an agricultural field, the optimized parameters were used to estimate crop water requirements from both the models. Excluding irrigation, rainfall and other fluxes were applied as boundary conditions. For estimating crop water requirement, inverse simulations at daily time steps were conducted, and soil moisture at surface (10 cm) and in the root zone was observed. A threshold soil moisture level was defined with 45% of management allowable depletion, triggering irrigation when moisture dropped below this value.

Figure S14 shows the optimized water requirements for crops using the SPM and DPM frameworks. The total water required under the DPM framework was 30 cm, whereas the SPM framework estimated a requirement of 24.6 cm, with both approaches involving four irrigation events. The

water needed to replenish soil moisture to field capacity was found to be 21% higher when considering the system as dual permeable. Additionally, irrigation estimated using the DPM framework closely aligns with the actual field irrigation applied, which was 28.47 cm.

The use of DPM over SPM indicates better representation of soil moisture dynamics in the subsurface. The results of this study may have implications in land-surface and hydrological modeling studies where subsurface modeling is an important component and agriculture is recognized as a separate land type.

6 | CONCLUSIONS

This study developed a GA-based inverse modeling framework for estimating DPM parameters using soil moisture observations at different temporal scales and at various depths. The effects of boundary condition and parameter combination on optimization results were analyzed using numerical and experimental data. In addition, the advantages of using a DPM in comparison to a single porosity model for predicting soil moisture, water fluxes, and crop water requirements were discussed.

The results highlighted that matrix parameters were generally easily identifiable with the exception of K_{sm} . The RZSM data at daily resolution provided the most reliable parameter estimates even under highly dynamic and variable boundary conditions, effectively capturing preferential flow signatures, offering an advantage over weekly data. Optimizing K_{as} along with other parameters increased the estimation accuracy of macropore parameters but compromised the accuracy of ω_f . The ω_f emerged as the key factor influencing macropore parametrization, revealing interdependence of parameters and underscoring the value of its prior knowledge. The framework also proved capable of handling uncertainty in soil moisture data; compared to SPM, DPM offered better prediction of soil moisture, fluxes within the root zone, and crop water requirement, thereby facilitating more accurate and precise assessment of water dynamics in agricultural fields.

While the framework demonstrated strong potential, certain limitations were also identified, like considering a homogeneous soil profile, simplifying assumption of constant initial condition, and reliance on soil moisture data alone. Future work should incorporate complementary data sources like ET and ground water flux, adopt finer temporal observations where available, and explore alternate optimization and machine learning techniques. Expanding the method to diverse soil and crop types can further validate its broader applicability.

Overall, the study confirms the effectiveness of GA-based inverse modeling in parametrizing DPM and provides prac-

tical insights for improving irrigation practices and water management under varying field conditions.

AUTHOR CONTRIBUTIONS

Nirali Vashishth: Conceptualization; data curation; formal analysis; investigation; methodology; writing—original draft. **Anoop Pandey:** Data curation; formal analysis; investigation; methodology; writing—original draft. **Richa Ojha:** Conceptualization; methodology; resources; software; supervision; validation; writing—review and editing.

ACKNOWLEDGMENTS

The first author is thankful to Ministry of Earth Sciences (MoES), Government of India, through grant number MoES/PAMC/H&C/116/2018-PC-II for funding this study. The second author acknowledges the Ministry of Education, GOI, for PMRF doctoral funding.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

ORCID

Nirali Vashishth  <https://orcid.org/0009-0005-1560-1751>

Anoop Pandey  <https://orcid.org/0009-0006-8834-5792>

Richa Ojha  <https://orcid.org/0000-0001-7014-8834>

REFERENCES

- Abu-hashim, M., Lilienthal, H., Schnug, E., Lasaponara, R., & Mohamed, E. S. (2023). Can a change in agriculture management practice improve soil physical properties. *Sustainability*, 15(4), 3573. <https://doi.org/10.3390/su15043573>
- Allen, R. G., Pereira, L. S., Raes, D., Smith, M., & Ab, W. (1998). *Yearbook of the United Nations*. Food and Agriculture Organization of the United Nations (FAO). <https://doi.org/10.18356/fa70fd58-en>
- Arora, B., Mohanty, B. P., & McGuire, J. T. (2011). Inverse estimation of parameters for multidomain flow models in soil columns with different macropore densities. *Water Resources Research*, 47(4), 1–17. <https://doi.org/10.1029/2010WR009451>
- Bandai, T., & Ghezzehei, T. A. (2022). Forward and inverse modeling of water flow in unsaturated soils with discontinuous hydraulic conductivities using physics-informed neural networks with domain decomposition. *Hydrology and Earth System Sciences*, 26(16), 4469–4495. <https://doi.org/10.5194/hess-26-4469-2022>
- Beven, K., & Germann, P. (2013). Macropores and water flow in soils revisited. *Water Resources Research*, 49(6), 3071–3092. <https://doi.org/10.1002/wrcr.20156>
- Cameira, M. R., Fernando, R. M., & Pereira, L. S. (2003). Soil macropore dynamics affected by tillage and irrigation for a silty loam alluvial soil in southern Portugal. *Soil & Tillage Research*, 70(2), 131–140. [https://doi.org/10.1016/S0167-1987\(02\)00154-X](https://doi.org/10.1016/S0167-1987(02)00154-X)
- Charoenhirunyingsos, S., Honda, K., Kamthongkiet, D., & Ines, A. V. M. (2011). Soil moisture estimation from inverse modeling using multiple criteria functions. *Computers and Electronics in Agriculture*, 75(2), 278–287. <https://doi.org/10.1016/j.compag.2010.12.004>
- Chen, C., Thomas, D. M., Green, R. E., & Wagenet, R. J. (1993). Two-domain estimation of hydraulic properties in macropore soils. *Soil*

- Science Society of America Journal*, 57(3), 680–686. <https://doi.org/10.2136/sssaj1993.03615995005700030008x>
- Choi, E. S., Cheema, T., & Islam, M. R. (1997). A new dual-porosity/dual-permeability model with non-Darcian flow through fractures. *Journal of Petroleum Science and Engineering*, 17(3–4), 331–344. [https://doi.org/10.1016/S0920-4105\(96\)00050-2](https://doi.org/10.1016/S0920-4105(96)00050-2)
- Cullum, R. F. (2009). Macropore flow estimations under no-till and till systems. *Catena*, 78(1), 87–91. <https://doi.org/10.1016/j.catena.2009.03.004>
- Das, N. N., & Mohanty, B. P. (2006). Root zone soil moisture assessment using remote sensing and vadose zone modeling. *Vadose Zone Journal*, 5(1), 296–307. <https://doi.org/10.2136/vzj2005.0033>
- Djabelkhir, K., Lauvernet, C., Kraft, P., & Carlier, N. (2017). Development of a dual permeability model within a hydrological catchment modeling framework: 1D application. *Science of the Total Environment*, 575, 1429–1437. <https://doi.org/10.1016/j.scitotenv.2016.10.012>
- Dorigo, W., Wagner, W., Albergel, C., Albrecht, F., Balsamo, G., Brocca, L., Chung, D., Ertl, M., Forkel, M., Gruber, A., Haas, E., Hamer, P. D., Hirschi, M., Ikonen, J., de Jeu, R., Kidd, R., Lahoz, W., Liu, Y. Y., Miralles, D., ... Lecomte, P. (2017). ESA CCI soil moisture for improved earth system understanding: State-of-the art and future directions. *Remote Sensing of Environment*, 203, 185–215. <https://doi.org/10.1016/j.rse.2017.07.001>
- Dorigo, W. A., Wagner, W., Hohensinn, R., Hahn, S., Paulik, C., Xaver, A., Gruber, A., Drusch, M., Mecklenburg, S., Van Oevelen, P., Robock, A., & Jackson, T. (2011). The international soil moisture network: A data hosting facility for global in situ soil moisture measurements. *Hydrology and Earth System Sciences*, 15(5), 1675–1698. <https://doi.org/10.5194/hess-15-1675-2011>
- Fan, J., Mcconkey, B., Wang, H., & Janzen, H. (2016). Field crops research root distribution by depth for temperate agricultural crops. *Field Crops Research*, 189, 68–74. <https://doi.org/10.1016/j.fcr.2016.02.013>
- Faúndez Urbina, C. A., van Dam, J., Tang, D., Gooren, H., & Ritsema, C. (2021). Estimating macropore parameters for HYDRUS using a meta-model. *European Journal of Soil Science*, 72(5), 2006–2019. <https://doi.org/10.1111/ejss.13103>
- Franklin, S. M., Kravchenko, A. N., Vargas, R., Vasilas, B., Fuhrmann, J. J., & Jin, Y. (2021). The unexplored role of preferential flow in soil carbon dynamics. *Soil Biology & Biochemistry*, 161, 108398. <https://doi.org/10.1016/j.soilbio.2021.108398>
- Fuhrmann, I., Maarastawi, S., Neumann, J., Amelung, W., Frindte, K., Knief, C., Lehdorff, E., Wassmann, R., & Siemens, J. (2019). Preferential flow pathways in paddy rice soils as hot spots for nutrient cycling. *Geoderma*, 337, 594–606. <https://doi.org/10.1016/j.geoderma.2018.10.011>
- Gad, A. F. (2023). PyGAD: An intuitive genetic algorithm Python library. *Multimedia Tools and Applications*, 83, 58029–58042. <https://doi.org/10.1007/s11042-023-17167-y>
- Garg, K. K., Das, B. S., Safeeq, M., & Bhadoria, P. B. S. (2009). Measurement and modeling of soil water regime in a lowland paddy field showing preferential transport. *Agricultural Water Management*, 96(12), 1705–1714. <https://doi.org/10.1016/j.agwat.2009.06.018>
- Gerke, H. H., & van Genuchten, M. T. (1993a). A dual-porosity model for simulating the preferential movement of water and solutes in structured porous media. *Water Resources Research*, 29(2), 305–319. <https://doi.org/10.1029/92WR02339>
- Gerke, H. H., & van Genuchten, M. T. (1993b). Evaluation of a first-order water transfer term for variably saturated dual-porosity flow models. *Water Resources Research*, 29(4), 1225–1238. <https://doi.org/10.1029/92WR02467>
- Gerke, H. H., & van Genuchten, M. T. (1996). Macroscopic representation of structural geometry for simulating water and solute movement in dual-porosity media. *Advances in Water Resources*, 19(6), 343–357. [https://doi.org/10.1016/0309-1708\(96\)00012-7](https://doi.org/10.1016/0309-1708(96)00012-7)
- Germann, P. F. (1990). Preferential flow and the generation of runoff: 1. Boundary layer flow theory. *Water Resources Research*, 26(12), 3055–3063. <https://doi.org/10.1029/WR026i012p03055>
- Goldberg, D. E. (1991). Genetic algorithms as a computational theory of conceptual design. In G. Rzevski & R. A. Adey (Eds.), *Applications of artificial intelligence in engineering VI* (pp. 3–16). Springer. https://doi.org/10.1007/978-94-011-3648-8_1
- Ines, A. V. M., & Droogers, P. (2002). Inverse modelling in estimating soil hydraulic functions: A genetic algorithm approach. *Hydrology and Earth System Sciences*, 6(1), 49–65. <https://doi.org/10.5194/hess-6-49-2002>
- Ines, A. V. M., & Mohanty, B. P. (2008). Near-surface soil moisture assimilation for quantifying effective soil hydraulic properties using genetic algorithm: 1. Conceptual modeling. *Water Resources Research*, 44(6), 1–26. <https://doi.org/10.1029/2007WR005990>
- Ines, A. V. M., & Mohanty, B. P. (2009). Near-surface soil moisture assimilation for quantifying effective soil hydraulic properties using genetic algorithms: 2. Using airborne remote sensing during SGP97 and SMEX02. *Water Resources Research*, 45(1), 1–22. <https://doi.org/10.1029/2008WR007022>
- Jarvis, N. J. (2020). A review of non-equilibrium water flow and solute transport in soil macropores: Principles, controlling factors and consequences for water quality. *European Journal of Soil Science*, 71(3), 279–302. <https://doi.org/10.1111/ejss.12973>
- Jarvis, N. J., Jansson, P.-E., Dik, P. E., & Messing, I. (1991). Modelling water and solute transport in macroporous soil. I. Model description and sensitivity analysis. *Journal of Soil Science*, 42(1), 59–70. <https://doi.org/10.1111/j.1365-2389.1991.tb00091.x>
- Kang, W., Tian, J., Reemt Bogena, H., Lai, Y., Xue, D., & He, C. (2023). Soil moisture observations and machine learning reveal preferential flow mechanisms in the Qilian Mountains. *Geoderma*, 438(222), 116626. <https://doi.org/10.1016/j.geoderma.2023.116626>
- Kodešová, R., Němeček, K., Jirků, V., Nikodem, A., FéA, M., Jakšík, O., Martin, K., & Žigová, A. (2010). Using the dye tracer experiment for characterisation of parameters of the dual-permeability model. In *Proceedings of the 19th World Congress of Soil Science, Soil Solutions for a Changing World* (pp. 117–120). International Union of Soil Sciences. <http://www.iuss.org/19thWCSS/Symposium/pdf/0133.pdf>
- Kodešová, R., Němeček, K., Kodeš, V., & Žigová, A. (2012). Using dye tracer for visualization of preferential flow at macro- and microscales. *Vadose Zone Journal*, 11(1), <https://doi.org/10.2136/vzj2011.0088>
- Köhne, J. M., Köhne, S., & Gerke, H. H. (2002). Estimating the hydraulic functions of dual-permeability models from bulk soil data. *Water Resources Research*, 38(7), 26–1–26–11. <https://doi.org/10.1029/2001wr000492>
- Köhne, J. M., Köhne, S., & Šimůnek, J. (2009). A review of model applications for structured soils: A) Water flow and tracer transport. *Journal of Contaminant Hydrology*, 104(1–4), 4–35. <https://doi.org/10.1016/j.jconhyd.2008.10.002>
- Köhne, J. M., & Mohanty, B. P. (2005). Water flow processes in a soil column with a cylindrical macropore: Experiment and hierarchical

- modeling. *Water Resources Research*, 41(3), 1–17. <https://doi.org/10.1029/2004WR003303>
- Köhne, J. M., Mohanty, B. P., & Šimůnek, J. (2006). Inverse dual-permeability modeling of preferential water flow in a soil column and implications for field-scale solute transport. *Vadose Zone Journal*, 5(1), 59–76. <https://doi.org/10.2136/vzj2005.0008>
- Kulasekera, P. B., Parkin, G. W., & von Bertoldi, P. (2011). Using soil water content sensors to characterize tillage effects on preferential flow. *Vadose Zone Journal*, 10(2), 683–696. <https://doi.org/10.2136/vzj2010.0063>
- Kumar, R., Shankar, V., & Jat, M. K. (2015). Evaluation of root water uptake models—A review. *ISH Journal of Hydraulic Engineering*, 21(2), 115–124. <https://doi.org/10.1080/09715010.2014.981955>
- Larsbo, M., Roullet, S., Stenemo, F., Kasteel, R., & Jarvis, N. (2005). An improved dual-permeability model of water flow and solute transport in the vadose zone. *Vadose Zone Journal*, 4(2), 398–406. <https://doi.org/10.2136/vzj2004.0137>
- Lennartz, B., Michaelsen, J., Wichtmann, W., & Widmoser, P. (1999). Time variance analysis of preferential solute movement at a tile-drained field site. *Soil Science Society of America Journal*, 63(1), 39–47. <https://doi.org/10.2136/sssaj1999.03615995006300010007x>
- Liu, H., Janssen, M., & Lennartz, B. (2016). Changes in flow and transport patterns in fen peat following soil degradation. *European Journal of Soil Science*, 67(6), 763–772. <https://doi.org/10.1111/ejss.12380>
- Luo, L., Lin, H., & Halleck, P. (2008). Quantifying soil structure and preferential flow in intact soil using X-ray computed tomography. *Soil Science Society of America Journal*, 72(4), 1058–1069. <https://doi.org/10.2136/sssaj2007.0179>
- Mitchell, M. (1998). *An introduction to genetic algorithms*. MIT Press.
- Mualem, Y. (1976). A new model for predicting the hydraulic conductivity of unsaturated porous media. *Water Resources Research*, 12(3), 513–522. <https://doi.org/10.1029/WR012i003p00513>
- Nimmo, J. R. (2010). Theory for source-responsive and free-surface film modeling of unsaturated flow. *Vadose Zone Journal*, 9(2), 295–306. <https://doi.org/10.2136/vzj2009.0085>
- Nimmo, J. R. (2012). Preferential flow occurs in unsaturated conditions. *Hydrological Processes*, 26(5), 786–789. <https://doi.org/10.1002/hyp.8380>
- Nimmo, J. R. (2021). The processes of preferential flow in the unsaturated zone. *Soil Science Society of America Journal*, 85(1), 1–27. <https://doi.org/10.1002/saj2.20143>
- Nimmo, J. R., & Mitchell, L. (2013). Predicting vertically nonsequential wetting patterns with a source-responsive model. *Vadose Zone Journal*, 12(4), vzj2013.03.0054. <https://doi.org/10.2136/vzj2013.03.0054>
- Nimmo, J. R., Wickenkamp, I., Araki, R., Groh, J., Singh, N. K., Crompton, O., Wyatt, B. M., Ajami, H., Giménez, D., Hirmas, D. R., Sullivan, P. L., & Sprenger, M. (2025). Identifying preferential flow from soil moisture time series: Review of methodologies. *Vadose Zone Journal*, 24(2), 1–25. <https://doi.org/10.1002/vzj2.70017>
- Nocedal, J., & Wright, S. J. (2006). Numerical optimization. In *Springer series in operations research and financial engineering*. <https://doi.org/10.1007/978-0-387-40065-5>
- Oberdörster, C., Vanderborght, J., Kemna, A., & Vereecken, H. (2010). Investigating preferential flow processes in a forest soil using time domain reflectometry and electrical resistivity tomography. *Vadose Zone Journal*, 9(2), 350–361. <https://doi.org/10.2136/vzj2009.0073>
- Ochsner, T. E., Cosh, M. H., Cuenca, R. H., Dorigo, W. A., Draper, C. S., Hagimoto, Y., Kerr, Y. H., Larson, K. M., Njoku, E. G., Small, E. E., & Zreda, M. (2013). State of the art in large-scale soil moisture monitoring. *Soil Science Society of America Journal*, 77(6), 1888–1919. <https://doi.org/10.2136/sssaj2013.03.0093>
- Ojha, C. S. P., & Rai, A. K. (1996). Nonlinear root-water uptake model. *Journal of Irrigation and Drainage Engineering*, 122(4), 198–202. [https://doi.org/10.1061/\(asce\)10733-9437\(1996\)122:4\(198\)](https://doi.org/10.1061/(asce)10733-9437(1996)122:4(198))
- Pandey, A., & Ojha, R. (2024). Macropore flow and its characteristics. In K. K. Singh & C. S. Prasad Ojha (Eds.), *Sustainable management of land, water and pollution of built-up area* (pp. 111–133). Springer. https://doi.org/10.1007/978-3-031-56176-4_9
- Panwar, A. S., Shamim, M., Babu, S., Ravishankar, N., Prusty, A. K., Alam, N. M., Singh, D. K., Bindhu, J. S., Kaur, J., Dashora, L. N., Pasha, M. D. L., Chatterjee, S., Sanjay, M. T., & Desai, L. J. (2018). Enhancement in productivity, nutrients use efficiency, and economics of rice-wheat cropping systems in India through farmer's participatory approach. *Sustainability*, 11(1), 122. <https://doi.org/10.3390/su11010122>
- Peters, A., & Durner, W. (2008). Simplified evaporation method for determining soil hydraulic properties. *Journal of Hydrology*, 356(1–2), 147–162. <https://doi.org/10.1016/j.jhydrol.2008.04.016>
- Pluer, W. T., Macrae, M., Buckley, A., & Reid, K. (2020). Contribution of preferential flow to tile drainage varies spatially and temporally. *Vadose Zone Journal*, 19(1), 1–15. <https://doi.org/10.1002/vzj2.20043>
- Raimbault, J., Peyneau, P. E., Courtier-Murias, D., Bigot, T., Gil Roca, J., Béchet, B., & Lassabatère, L. (2021). Investigating the impact of exit effects on solute transport in macroporous media. *Hydrology and Earth System Sciences*, 25(2), 671–683. <https://doi.org/10.5194/hess-25-671-2021>
- Ramos, T. B., Gonçalves, M. C., Martins, J. C., Genuchten, M. T., & Pires, F. P. (2006). Estimation of soil hydraulic properties from numerical inversion of tension disk infiltrometer data. *Vadose Zone Journal*, 5(2), 684–696. <https://doi.org/10.2136/vzj2005.0076>
- Salonga, P. K., Inaudito, J. M., & Mendoza, R. (2019). An unconstrained minimization technique using successive implementations of Golden Search algorithm. In *AIP Conference Proceedings* (Vol. 2192, No. 1, p. 060018). AIP Publishing LLC. <https://doi.org/10.1063/1.5139164>
- Shankar, V., Hari Prasad, K. S., Ojha, C. S. P., & Govindaraju, R. S. (2012). Model for nonlinear root water uptake parameter. *Journal of Irrigation and Drainage Engineering*, 138(10), 905–917. [https://doi.org/10.1061/\(asce\)ir.1943-4774.0000469](https://doi.org/10.1061/(asce)ir.1943-4774.0000469)
- Šimůnek, J., Jarvis, N. J., van Genuchten, M. T., & Gärdenäs, A. (2003). Review and comparison of models for describing non-equilibrium and preferential flow and transport in the vadose zone. *Journal of Hydrology*, 272(1–4), 14–35. [https://doi.org/10.1016/S0022-1694\(02\)00252-4](https://doi.org/10.1016/S0022-1694(02)00252-4)
- Šimůnek, J., Šejna, M. A., Saito, H., Sakai, M., & Van Genuchten, M. T. (2013). *The HYDRUS-1D software package for simulating the movement of water, heat, and multiple solutes in variably saturated media, version 4.17*. HYDRUS Software Series 3D, Department of Environmental Sciences, University of California Riverside. https://www.pc-progress.com/Downloads/Pgm_Hydrus1D/HYDRUS1D-4.17.pdf
- Simunek, J., & van Genuchten, M. T. (1999). Using the HYDRUS-1D and HYDRUS-2D codes for estimating unsaturated soil hydraulic and solute transport parameters. *Characterization and Measurement of the Hydraulic Properties of Unsaturated Porous Media*, 2, 1523–1536. https://www.ars.usda.gov/arsuserfiles/20360500/pdf_pubs/P1667.pdf

- Sonkar, I., Kaushika, G. S., & Hari Prasad, K. S. (2018). Modeling moisture flow in root zone: Identification of soil hydraulic and root water uptake parameters. *Journal of Irrigation and Drainage Engineering*, 144(10), 1–11. [https://doi.org/10.1061/\(asce\)ir.1943-4774.0001342](https://doi.org/10.1061/(asce)ir.1943-4774.0001342)
- Sonkar, I., Kotnoor, H. P., & Sen, S. (2019). Estimation of root water uptake and soil hydraulic parameters from root zone soil moisture and deep percolation. *Agricultural Water Management*, 222, 38–47. <https://doi.org/10.1016/j.agwat.2019.05.037>
- Soracco, C. G., Lozano, L. A., Sarli, G. O., Gelati, P. R., & Filgueira, R. R. (2011). Using tension disc infiltrometer to determine infiltration and water-conducting macroporosity and mesoporosity relationships in an agricultural silty loam soil. *Soil Science*, 176(9), 459–463. <https://doi.org/10.1097/SS.0b013e3182257d65>
- Talukder, R., Plaza-Bonilla, D., Cantero-Martínez, C., Wendroth, O., & Lampurlanés, J. (2023). Soil hydraulic properties and pore dynamics under different tillage and irrigated crop sequences. *Geoderma*, 430, 116293. <https://doi.org/10.1016/j.geoderma.2022.116293>
- Valdes-Abellan, J., Pachepsky, Y., & Martinez, G. (2018). Obtaining soil hydraulic parameters from soil water content data assimilation under different climatic/soil conditions. *Catena*, 163, 311–320. <https://doi.org/10.1016/j.catena.2017.12.022>
- van Genuchten, M. T. (1980). A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Science Society of America Journal*, 44(5), 892–898. <https://doi.org/10.2136/sssaj1980.03615995004400050002x>
- van Genuchten, M. T., & Wierenga, P. J. (1976). Mass transfer studies in sorbing porous media I. Analytical solutions. *Soil Science Society of America Journal*, 40(4), 473–480. <https://doi.org/10.2136/sssaj1976.03615995004000040011x>
- Walker, J. P., Willgoose, G. R., & Kalma, J. D. (2004). In situ measurement of soil moisture: A comparison of techniques. *Journal of Hydrology*, 293(1–4), 85–99. <https://doi.org/10.1016/j.jhydrol.2004.01.008>
- Wang, D., Yates, S. R., & Ernst, F. F. (1998). Determining soil hydraulic properties using tension infiltrometers, time domain reflectometry, and tensiometers. *Soil Science Society of America Journal*, 62(2), 318–325. <https://doi.org/10.2136/sssaj1998.03615995006200020004x>
- Yu, D., Yang, J., Shi, L., Zhang, Q., Huang, K., Fang, Y., & Zha, Y. (2019). On the uncertainty of initial condition and initialization approaches in variably saturated flow modeling. *Hydrology and Earth System Sciences*, 23(7), 2897–2914. <https://doi.org/10.5194/hess-23-2897-2019>

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Vashishth, N., Pandey, A., & Ojha, R. (2025). Estimation of dual permeability model parameters in agricultural fields using a genetic algorithm-based inverse modeling framework. *Vadose Zone Journal*, 24, e70067. <https://doi.org/10.1002/vzj2.70067>